

The Price of a Design Parameter

Bank Valuations in the Digital Euro Legislative Process

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Abstract

The holding limit on the digital euro was designed to prevent deposit flight from banks. We ask whether markets priced the design choices legislators made between 2023 and 2026. We code 22 verified event dates by their available design content before looking at any price, and add the published record of 309 disclosed lobbying meetings. A euro area bank one standard deviation more deposit-funded gains 0.05 percentage points more on an event that limits the digital euro (permutation $p = 0.45$). Conservative release-time alignment raises this to 0.15–0.17, some evidence of a small effect that is not robust across inference procedures; every pooled specification excludes effects one third the size of those documented for 2020. Events that set a design parameter show no detectable effect. The same design, with strictly pre-event betas, detects an effect close to the 2020 size on the day the Eurosystem report appeared. Markets appear to have reacted to early news about whether a digital euro would exist; we find no comparable response to later news about its design.

Keywords: central bank digital currency; digital euro; financial stability; deposit flight; payments; event study; political economy of regulation.

JEL classification: E42, E58, G21, G28, D72.

1 Introduction

The central financial-stability concern about a retail central bank digital currency is deposit flight: households moving balances out of banks into central bank money, gradually or in a run. The holding limit on the digital euro exists to contain that risk. Whether it does so, and whether markets believe it does, depends on how the instrument is designed, and the common description of a CBDC as a simple threat to bank deposits is incomplete in a way that matters for policy. The digital euro, as it has actually been drafted, is a bundle of parameters: a ceiling on how much any person may hold, a rule on whether firms may hold at all, a cap on what merchants and payment providers may charge each other, a mandate that device manufacturers open their hardware, and a decision on whether an online version is built at all. Each parameter moves rent from one part of the payments chain to another. The aggregate effect on banks is the sum of these transfers, and it can be positive or negative depending on how the parameters are set.

This distinction is not academic. Between June 2023 and July 2026 the European Parliament and the Council did not debate whether to have a digital euro. They debated the parameters. The rapporteur's draft report of November 2025 proposed that the online version be built only if the Commission first concluded that no private pan-European alternative exists. The Council's position of December 2025 capped interchange and merchant fees for at least five years and required device manufacturers to grant access to their hardware. The committee position of June 2026 barred firms from holding digital euro beyond twenty-four hours. Each of these is a transfer with a price, and the prices are not known.

This paper. We measure what a design parameter is worth, and to whom. The empirical design is a two-stage event study over the legislative phase. In the first stage we compute abnormal returns for the 49 listed firms with usable price data, 29 of them euro area banks, from factor models appropriate to the market of listing. In the second stage we relate them to firm exposure interacted with the coded design content of each event, with firm and event fixed effects, so that everything common to a trading day is absorbed and identification comes from the cross-section.

What makes the design work. The events are not one-dimensional. The same announcement can be good news for banks and bad news for payment providers, because it limits quantities and caps fees at the same time. Figure 1 and Table 2 show the coded pattern: the committee position and its confirmation in plenary carry opposite signs for banks and payment providers, and several other events move one group while leaving another untouched. A common news-intensity channel cannot produce an announcement that raises one group and lowers another. This is the feature that separates our design from an event study of project news, and it is the reason we code content before seeing prices.

A second, independent source. The European Parliament publishes every meeting between rapporteurs, shadow rapporteurs, committee chairs and interest representatives on the procedure file. We transcribe the complete published record: 309 disclosed meetings between July 2023 and July 2026, with date, member, role and organisation. Deutsche Bank appears twelve times, the French banking federation ten, Amazon and the European consumer organisation eight each, Banco Santander seven.

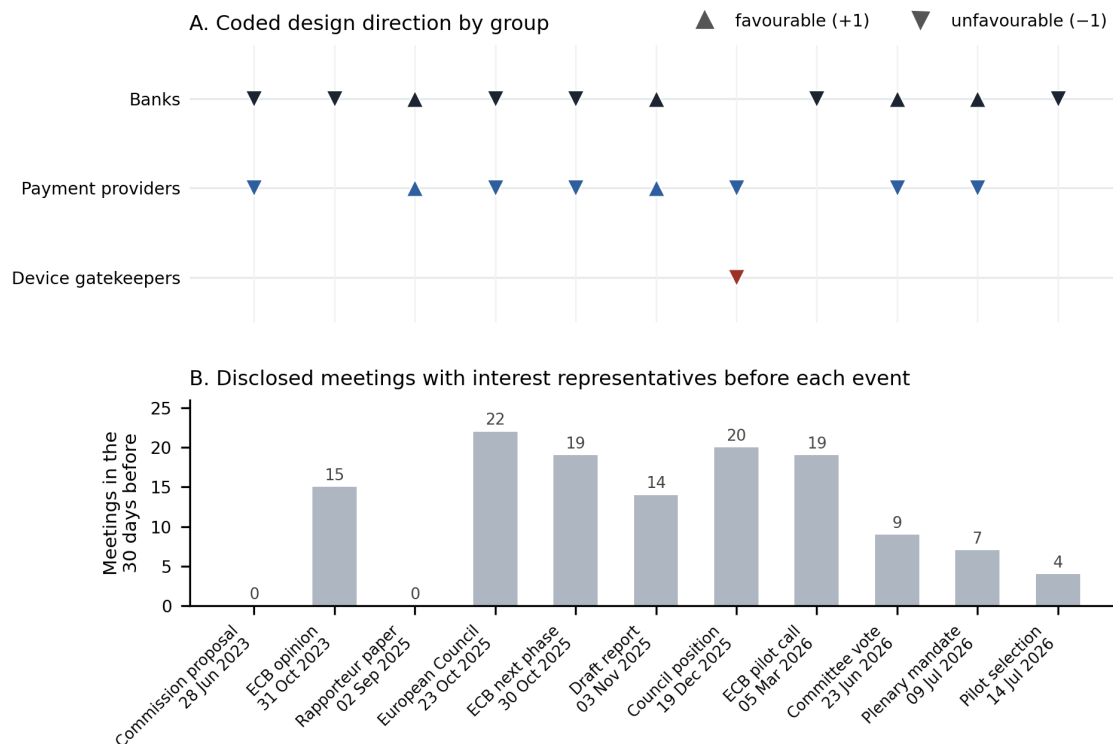


Figure 1: The digital euro file: contestation and coded design content

Notes: Events with a non-zero coded direction for at least one group, in chronological order. Direction is coded from the contemporaneously available text of the act before any price data was loaded; groups coded zero for an event are left blank. Panel B counts meetings disclosed on procedure file 2023/0212(COD) by rapporteurs, shadow rapporteurs and the committee chair in the 30 calendar days before each event.

This record serves three purposes. It documents which interests bargained over which parameter. It provides a measure, observable before the event, of how contested each step was, which speaks to the anticipation critique that event studies of legislation usually face. And it validates the falsification events from outside the price data: exactly one meeting precedes the purely procedural event of November 2024, against twenty-two in the thirty days before the Eurogroup agreement on the holding-limit mechanism.

What we find. The legislative design of the digital euro moved bank valuations little, if at all. In the pre-specified specification, a euro area bank one standard deviation more deposit-funded gains 0.05 percentage points more on an event that limits the digital euro, with a permutation p -value of 0.45. The confidence interval excludes effects one sixth the size of those that Burlon et al. (2024) document for the ECB's announcements of 2020 and 2021. A conservative alignment of events to their release time, which uses two-day windows for the thirteen releases whose time of day is not documented, raises the estimate to between 0.15 and 0.17 percentage points depending on the return benchmark, with permutation p -values between 0.05 and 0.12; under a fully euro-denominated benchmark the wild cluster bootstrap p -value is 0.02. That is some evidence of a small effect under conservative timing, not a robust finding. Every pooled specification we run excludes effects one third the size of the earlier ones. In a decomposition we added after the pre-specified analysis, the events that fixed

a design parameter, as distinct from events that only changed the likelihood of issuance, show no detectable effect in any event window. Contested and unscheduled events show nothing, which speaks against anticipation, and the same exposure slope appears for banks outside the euro area, which is consistent with a generic factor loading on deposit-funded banks rather than a digital-euro-specific response. Payment providers show no consistent response, and the response of device gatekeepers rests on a single event; our evidence on the non-bank side of the payments chain is correspondingly weaker than on banks, and the paper's empirical claims concern banks first.

The null is not a failure of the design to see anything. Extending the sample to 2020 and 2021 with the same firms, exposure measure and estimator, the design detects an effect of 0.85 percentage points per standard deviation, with exposure measured before the event, on the day the Eurosystem published its report on a digital euro, close to the magnitude documented in the earlier literature. What it does not detect, in 2021 or later, is any response to news about how the instrument would be designed, including the first signal of a holding limit in February 2021. The market appears to have reacted to early news about whether a digital euro would exist; we find no comparable response to news about its design.

Why it matters. Two conclusions follow for policy. For financial stability, markets did not assign a large residual deposit-flight risk to euro area banks during the legislative process: any effect on banks with very different exposure appears modest relative to the response to the early digital euro announcements. The parameters that took three years to negotiate were, in the market's assessment, second order to bank value. This is consistent with the simulation evidence of Meller and Soons (2023): the limits under discussion, up to EUR 3,000 per person, are within the range for which they find contained effects on bank liquidity and funding. For the political economy of the file, the negotiation record shows that the same parameters were intensely contested, with the banking side the most frequent counterparty of the legislators. The contrast between what was bargained over and what was priced is itself informative: whatever the banks were negotiating for did not generate a large detectable valuation response in equity markets.

Related literature. The paper connects to five literatures.

CBDC and bank intermediation. The canonical concern is that a central bank liability available to households substitutes for deposits and shrinks the funding base of banks. Brunnermeier and Niepelt (2019) show conditions under which the substitution is neutral, because the central bank can pass the funds back to banks; Fernández-Villaverde et al. (2021) show that a CBDC may concentrate deposits at the central bank in a run; Andolfatto (2021) shows that a CBDC can raise deposits when banks have market power. Keister and Sanches (2023) formalise the trade-off between more efficient exchange and crowded-out deposits, with an optimum that depends on how valuable bank intermediation is at the margin. Chiu et al. (2023) add imperfect competition and overturn the sign: with deposit market power, which is large in the data (Drechsler et al. 2017), a CBDC forces deposit rates up and can *raise* deposits and lending. The empirical question is therefore not whether a CBDC hurts banks but where in the cross-section it hurts and by how much. Our design speaks directly to the cross-sectional question and only indirectly to the level.

Quantity limits as the policy instrument. Because the sign of the intermediation effect is

ambiguous, the policy debate has concentrated on bounding the quantity. Bidder et al. (2024) distinguish slow deposit substitution in normal times from fast disintermediation during banking stress and show in their calibrated model that a suitably chosen holding limit can contain the run-risk channel. The European negotiation uses a standing ceiling reviewed on a fixed cycle, so our events speak to the quantity channel rather than directly to run dynamics. For the euro area, Meller and Soons (2023) simulate a digital euro with supervisory data on more than 2,000 banks and find that a EUR 3,000 per-person holding limit would have contained effects on bank liquidity risks and funding structures even in extremely pessimistic scenarios. Lambert et al. (2024) likewise assess safeguards designed to limit effects on bank liquidity and central-bank reliance. Muñoz and Soons (2023) study the demand for public money as a store of value when beliefs are heterogeneous. Auer et al. (2025) provide further simulation evidence on the banking system, and Chapman et al. (2023) survey the literature on CBDC and banking. This work is model-based and simulates the balance-sheet consequences of given limits. Our contribution is the market's own assessment of the limits actually negotiated.

Event studies of CBDC news. Burlon et al. (2024) are closest to us and provide the benchmark for magnitudes. They show that news about the digital euro moved euro area bank valuations, scaled by each bank's deposit ratio, with more than one percentage point of abnormal return per standard deviation of the deposit ratio and event, and they connect the valuation effects to lending through credit register data. Their events are communications by members of the ECB's Executive Board in 2020 and 2021, and the sample ends in May 2021. They observe that the trend reverses after the Executive Board first floated a holding limit of EUR 3,000 in February 2021, which is the mechanism we make systematic. Berg et al. (2024) study digital euro announcements for payment firms: valuations of US payment firms fall and those of European payment firms rise, while bank stocks do not react significantly. We differ from both in three ways. Our events are legislative and cover the period in which the co-legislators adopted and narrowed their positions on the design parameters. We code each event by its design content for each group rather than treating it as homogeneous news. And we test incidence channel by channel: the fee channel applies two-sided-platform logic (Rochet and Tirole 2003) to a regulated new rail, and the device-access provisions operate one layer further down, at the hardware interface.

Political economy of financial regulation. Our negotiation record connects to work on how competing interest groups shape financial legislation (Kroszner and Stratmann 1998), on lobbying by financial institutions before the global financial crisis (Igan et al. 2012), and on what lobbyists supply to legislators (Bertrand et al. 2014). Our reading of the record is deliberately modest. It shows which interests were present and when, which is informative about the contestation of each parameter, but presence is chosen by the interested party and is therefore endogenous to expected exposure. We use the record to validate the coding and to measure contestation, not as a source of variation.

Event-study methods. We build on event studies of regulatory news (Schwert 1981; Binder 1985; MacKinlay 1997). With few effective clusters and cross-sectionally correlated abnormal returns (Kolari and Pynnönen 2010), we rely on permutation inference over exposure within groups and a null-imposed wild cluster bootstrap (Cameron et al. 2008), and we assess finite-sample size by simulation.

Contribution. We provide the first event-study use of the digital euro’s legislative process to construct parameter-specific design events and measure their incidence across the payments value chain. We also contribute two reusable public-data objects: a primary-source event list with documented design content, and the complete negotiation record of the procedure file.

2 Institutional background

2.1 What the digital euro is, as drafted

The Commission proposed the regulation establishing the digital euro on 28 June 2023, as document COM(2023) 0369. The instrument as drafted has four features that determine its economic incidence.

It is a liability of the central bank, distributed by intermediaries. Payment service providers open and service digital euro accounts; the ECB issues the claim. Banks therefore face the substitution of a funding instrument and simultaneously acquire a distribution role.

Holdings are capped. Individual holdings are subject to a limit whose level was left to be decided. At the request of the co-legislators, the ECB later analysed hypothetical limits of EUR 500 to EUR 3,000 under business-as-usual and flight-to-safety scenarios (European Central Bank 2025). Lambert et al. (2024) analyse how holding limits can contain effects on bank liquidity and central-bank funding. The limit is the single parameter that determines how much deposit substitution is possible, and it was the most heavily contested item in the file.

It carries no interest. Remuneration is excluded by design, which caps the attractiveness of the instrument as a store of value and distinguishes it from the wholesale CBDC literature.

Its fee structure is regulated. What merchants pay, and what providers may charge each other, is set in the regulation rather than by the market. This is where the incidence for payment providers is determined, and it is economically distinct from the deposit channel.

2.2 The legislative process as a sequence of parameter decisions

The file passed through the Parliament and the Council between 2023 and 2026. Table 2 lists the events we use. Three deserve description because they carry most of the design content.

The rapporteur’s draft report, 3 November 2025. After the change of rapporteur in December 2024, the new draft splits the instrument. The offline digital euro, a tokenised device-to-device instrument with cash-like properties, is introduced immediately. The online version, which is the deposit-substituting one, is introduced only if the Commission, after an investigation phase, concludes that no suitable pan-European sovereign retail payment solution exists. Holding limits are made more prescriptive. This is the most quantity-limiting proposal in the file, and it also makes the project conditional on the failure of private European alternatives.

The Council position, 19 December 2025. Limits are set by the ECB but must respect an overall ceiling agreed by the Council and reviewed at least every two years, which moves control of the parameter from the central bank to the legislators. Providers may not charge consumers for mandatory services. A framework guarantees providers access to the hardware and software of mobile device manufacturers. Interchange and merchant service charges are capped for a transitional period of at

least five years at levels based on comparable means of payment, and are cost-based thereafter.

The committee position, 23 June 2026, confirmed in plenary on 9 July 2026. An overall cap on holdings is to be set by the Commission on the ECB's recommendation and reviewed at least every two years. Legal persons may not hold digital euro as a general rule and may accumulate incoming payments for at most twenty-four hours. The rollout phase is to last at least twenty-four months. Merchant and inter-provider fees are capped, and offline payments are free of charge.

2.3 The Commission baseline

The proposal of 28 June 2023 sets the baseline against which the two positions in Table 1 are amendments. Five features of it matter for incidence.

Legal tender. The digital euro is granted legal tender status within the euro area, with Member States designating authorities to enforce acceptance. Acceptance is therefore not a commercial decision, which is what makes the fee provisions binding rather than indicative: merchants cannot decline the instrument and negotiate the price of accepting it.

Online and offline, not programmable. The instrument is available in both forms and is convertible at par with cash and with commercial bank money. Non-programmability was a condition set by finance ministers before the proposal and is preserved in it.

Distribution by payment service providers. All EU providers may distribute, which the Commission underpinned with a parallel regulation for providers incorporated outside the euro area (COM(2023)368) and with the revision of the payment services framework (COM(2023)366) that extends the definition of funds to retail central bank money. Distribution is an obligation for credit institutions in respect of their own customers rather than an opportunity they may decline.

Quantitative limits as a financial stability tool. The proposal states that unrestricted use as a store of value could endanger financial stability and reduce credit provision, and that the ECB may introduce limits, including quantitative limits on individual holdings and limits on conversion. The level is not fixed in the proposal. This is the origin of the parameter that the Parliament and the Council then fought over for three years, and it is why the file is a natural experiment in the pricing of a design choice rather than of a policy announcement.

A package, not a single act. The proposal came with a regulation on the legal tender of cash (COM(2023)364), which is why the Council's December 2025 position covers both the digital euro and the acceptance of cash. Events that touch the cash regulation affect retailers and cash handlers and not banks, which is a further source of cross-sectional variation we code but do not rely on.

A note on dating the Council position. The Council working document carries 17 December 2025 and the press release announcing the position carries 19 December 2025. We use the press release date, because that is when the substance became public, and report the alternative date in Table 10.

2.4 The two positions in detail

By July 2026 the two co-legislators had adopted positions that differ on every parameter that matters for incidence. Table 1 sets them side by side. Both are verified from the primary documents: the Council press release of 19 December 2025 and the committee report tabled on 26 June 2026 and

Table 1: The two co-legislator positions on the parameters that determine incidence

Parameter	Council, 19 December 2025	Parliament, 23 June / 9 July 2026
Who sets the holding limit	ECB sets limits, but must respect an overall ceiling agreed by the Council, reviewed at least every two years	Commission sets the overall cap on the ECB's recommendation, reviewed at least every two years
Holdings by legal persons	Not specified in the press release	As a rule not permitted; incoming payments may be held at most 24 hours before being swept
Consumer fees	Providers may not charge for mandatory services; added-value services may be priced	No account maintenance, inactivity or minimum balance fees for basic services; practices circumventing free basic services prohibited
Merchant and interchange fees	Capped for a transitional period of at least five years at levels based on comparable means of payment; cost-based thereafter	Capped; offline payments free of charge
Access to mobile devices	Framework guaranteeing providers access to the hardware and software of device manufacturers	Not the central instrument in the committee text
Rollout	Not specified in the press release	Rollout phase of at least 24 months after authorisation, preceded by pilot testing
Liability and privacy	Not specified in the press release	Users bear no losses from outages; the ECB has no access to personal identification data under any circumstances

Notes: Entries are taken verbatim in substance from the two primary documents. “Not specified in the press release” means the item is not covered in the source we verified and not that the position is silent; the consolidated texts will resolve these cells. The comparison is the object of the interinstitutional negotiations that followed the plenary mandate of 9 July 2026.

confirmed in plenary on 9 July 2026.

Three observations follow, and each maps to a channel in Section 4.

Control over the quantity parameter moved away from the central bank in both positions, but to different bodies. The Council reserves the ceiling to itself; the Parliament gives it to the Commission on the ECB's recommendation. Either way the parameter that determines deposit substitution became a political variable, reviewed on a two-year cycle. For banks this is favourable relative to unconstrained central bank discretion, because the ceiling is now set by bodies that are directly exposed to their representations, and the negotiation record shows that they made them.

The fee provisions bind providers, not banks. Capping merchant and inter-provider charges transfers rent to merchants. Banks are affected only through their distribution income, which is second order relative to the deposit channel. This is what makes the December 2025 and June 2026 acts two-sided.

The device-access provision has one addressee. Requiring manufacturers to open hardware and software interfaces to payment providers is directed at firms that control a mobile operating system. The record shows two meetings between the rapporteur or a shadow rapporteur and the largest such firm, in November 2025, three weeks before the Council fixed its position, and in June 2026, a week before the committee vote.

2.5 Why the market could price these events

Three conditions have to hold for an event study to be informative here, and all three are satisfied.

First, the acts are public on the day. Committee votes, plenary results, Council press releases and ECB publications are dated, and the substantive content is in the release rather than in a consolidated text published weeks later. The time of day is documented for some releases and not for others; Section 5.6 explains how we treat that.

Second, the amounts are large enough to matter for valuations. Euro area overnight deposits of households and non-financial corporations are of the order of several trillion euro, and a holding limit of EUR 3,000 applied to a few hundred million residents defines a substitution capacity in the high hundreds of billions. Whether take-up would approach that capacity is precisely the uncertainty the limit resolves.

Third, the affected firms are listed and the exposure is measurable from public reporting. That is not true of every regulatory file, and it is what makes this one usable.

2.6 The 2024 standstill and what it implies for coding

The file did not progress in 2024, and the reason is documented rather than inferred. The draft report tabled by the first rapporteur on 9 February 2024 was never put to a vote. In mid-2024 three shadow rapporteurs wrote to the committee chair complaining that the rapporteur had delayed the process from the start. In December 2024 the rapporteur stepped down and was replaced.

What this implies for coding requires care. In the pre-specified coding the draft of February 2024 and the amendments tabled on it were coded as carrying no design direction, and part of the justification was that the draft was never voted. That justification uses information the market did not have on the day: in February 2024 nobody knew that the draft would stall. The correct question is what a market participant could learn from the text on the day it was published. Because we were unable to retrieve the text of the draft, we cannot answer that question from its content, and the correct treatment is to exclude both events rather than to code them as neutral. We report the pre-specified estimate and the estimate without them; since both carried a zero direction, the change affects only the fixed effects and, as Section 9.8 shows, leaves the result unchanged.

The negotiation record is consistent with a file that stalled in 2024: documented meetings fall from 48 in 2023 to 20 in 2024 and rise to 161 in 2025. We use this only as description, not as a reason for any coding decision.

The change of rapporteur. At the date of appointment, December 2024, the successor had no publicly documented position on the file. The paper in which he argues that the digital euro is not the answer to European dependence on non-European card networks appeared only in 2025. We therefore code the appointment itself as carrying no direction, and treat the later publication of his position, on 2 September 2025, as a separate event. Coding the appointment as bank-favourable because of what the appointee later wrote would be using information the market did not have.

A note on dating the draft report. The draft report carries 3 November 2025 and was presented to the committee on 5 November 2025. We use the document date in the main specification and report the presentation date in Table 10, because we cannot establish from the record which came first in the

public domain.

2.7 The rapporteur's revealed position

One event in the sample is neither a legislative act nor a central bank decision. On 2 September 2025 the rapporteur published a twenty-seven page chapter in a yearbook on the euro, arguing that the digital euro is not needed and pointing instead to private European alternatives, naming the European Payments Initiative. He presented the piece as a personal contribution rather than as the rapporteur's position.

We include it and code it as favourable for banks and for private European payment schemes, for three reasons. The substance reappeared two months later in his draft report, where the online digital euro is made conditional on the absence of a private pan-European solution and the assessment is assigned to the Commission rather than to the ECB. The author is the single member with agenda control over the file. And the framing as personal does not change what a market participant learns from it, which is the direction in which the file will be steered.

The disclaimer is nevertheless recorded, and the event is included in the leave-one-out set so that the main estimate can be read with and without it.

2.8 The parallel operational track

The ECB ran its own timetable alongside the legislature. The preparation phase closed on 30 October 2025 and the project moved to its next phase. A call for expression of interest for pilot participants was published on 5 March 2026 and closed on 14 May 2026; on 14 July 2026 the ECB selected 36 payment service providers from more than 50 applicants, covering banks and non-banks. A call for merchants followed on 15 September 2026. A twelve-month pilot is planned for the second half of 2027, and first issuance is possible in 2029 if the regulation is adopted in 2026.

This track matters for identification in two ways. It produces events that advance the project without changing its design, which are useful as a contrast to design events. And the selection of pilot participants is, in principle, a within-group event: it separates providers that will participate from those that will not, on a single day.

3 Conceptual framework

We do not need a full model to sign the empirical predictions. What we need is an accounting of where rent sits in the payments chain and how each parameter moves it.

Consider three incumbent groups. Banks earn a spread s on their deposit base D_i , and they also earn fees from distributing payment services. Payment providers, including card networks and acquirers, earn a margin m on their transaction volume V_j . Device gatekeepers earn a rent r on each unit of mobile payment volume V^{mobile} routed through the hardware interface they control.

A digital euro with holding limit L , a rule $\lambda \in \{0, 1\}$ on whether legal persons may hold it, a fee cap $\bar{f}$, and a device-access mandate $a \in \{0, 1\}$ changes these as follows.

The holding limit is the deposit channel. Substitution away from deposits is bounded by L times

the number of users for whom the limit is not slack. A higher L therefore lowers bank rent, and by Proposition 1 the marginal loss is larger for banks with a larger fraction of depositors whose balances exceed the limit, since only those depositors can move more when the limit is raised. This is the channel Burlon et al. (2024) measure and the one Chiu et al. (2023) argue may reverse under imperfect competition, because the threat of substitution forces banks to raise deposit rates only where they previously exercised market power.

The rule for legal persons scales the same channel. Barring legal persons from holding removes corporate balances from the substitutable base entirely.

The fee cap is a separate channel that runs through providers, not banks. A binding cap $\bar{f}$ transfers rent from providers to merchants without touching deposits. It is possible, and in the committee position of June 2026 it is the case, for one act to limit the deposit channel and tighten the fee channel at the same time.

The device-access mandate is a third channel with a single addressee. Mandated access to hardware dissipates the gatekeeper's rent rV^{mobile} and passes part of it to providers.

Three predictions follow and are taken to the data.

- P1 Events that limit quantity raise bank valuations, and the effect scales with the share of deposits that are potentially substitutable.
- P2 Events that cap fees lower the valuations of payment providers, and the effect scales with the share of revenue earned in euro area transactions.
- P3 Events that mandate device access lower the valuation of gatekeepers and raise that of providers.

Prediction P3 is the sharpest, because the addressee is essentially unique. Predictions P1 and P2 are the ones that generate opposite signs on the same day, which is the feature the design rests on.

4 A model of rent incidence in the payments chain

The purpose of the model is narrow: to derive which group gains and which loses from each design parameter, so that the empirical coding is a prediction rather than a judgement call. It is a static model of a payments market with three incumbent groups and one entrant, the central bank.

4.1 Environment

A unit mass of households holds transaction balances b , distributed with cumulative distribution F on $[0, \bar{b}]$. Households make payments of total value V . There are three incumbents.

Banks take deposits and earn a spread. Bank i holds deposits $D_i = \int_0^{\bar{b}} b dF_i(b)$ and earns sD_i , where the spread s is the difference between the market rate and the deposit rate it pays. Banks differ in F_i : some have deposit bases concentrated in small transaction balances, others in large ones.

Payment providers process transactions. Provider j handles volume V_j and earns a margin m per unit, so revenue is mV_j . The margin has two components, an interchange fee received from the acquiring side and a merchant service charge.

Device gatekeepers control the hardware interface through which mobile payments are initiated.

A gatekeeper earns r per unit of volume routed through its interface, either as an explicit fee or as the shadow value of excluding competitors.

The central bank issues a digital euro characterised by four parameters:

- L the holding limit per person;
- λ an indicator, $\lambda \in \{0, 1\}$, for whether legal persons may hold;
- $\bar{f}$ a cap on the fees providers may charge;
- a an indicator, $a \in \{0, 1\}$, for whether device access is mandated.

4.2 Substitution and its bound

A household substitutes from deposits into digital euro up to $\min(b, L)$. Aggregate substitution from bank i is therefore

$$S_i(L) = \sigma \int_0^{\bar{b}} \min(b, L) dF_i(b), \quad (1)$$

where $\sigma \in [0, 1]$ is the take-up rate. Two properties of (1) drive everything that follows.

Proposition 1 (Concavity of substitution in the limit). *$S_i(L)$ is increasing and concave in L , with $S'_i(L) = \sigma [1 - F_i(L)]$. The marginal effect of raising the limit is therefore proportional to $1 - F_i(L)$, the fraction of the bank's depositors whose balances exceed L , and it falls to zero as L approaches $\bar{b}$.*

Proposition 1 has an immediate empirical implication that distinguishes our design from a linear one: the same increment in L is worth more when L is low, and worth more to a bank whose deposits sit in large balances. It also means that a cap set well above the typical transaction balance is close to no cap at all, which is why the debate between EUR 500 and EUR 3,000 was economically substantive rather than symbolic.

Proposition 2 (Exposure ordering). *Let banks i and k have deposit distributions with F_i first-order stochastically dominated by F_k . Then $S_i(L) \leq S_k(L)$ for every L . The level gap $S_k(L) - S_i(L)$ is weakly increasing in L , and the difference in marginal effects, $S'_k(L) - S'_i(L) = \sigma [F_i(L) - F_k(L)]$, is largest where the two distribution functions differ most.*

Two implications matter for measurement. The *level* of substitution a bank faces at a given limit rises with its substitutable deposit base, which motivates a deposit share as the exposure measure. The *marginal* effect of changing the limit depends on the fraction of depositors with balances above the limit, which would require account-level balance distributions that are not public. Our measures therefore capture the first implication directly and the second only by proxy.

4.3 The three channels

Bank i 's rent is $\Pi_i^B = s [D_i - S_i(L) - \lambda C_i] + \phi_i$, where $C_i \geq 0$ are the corporate deposit balances that would move into digital euro if legal persons were permitted to hold it, and ϕ_i is distribution income from servicing digital euro accounts, which is bounded by the fee regulation. Provider j 's rent is $\Pi_j^P = \min(m, \bar{f}) V_j + a \rho_j r V^{\text{mobile}} - c_j$, where $\rho_j \geq 0$, with $\sum_j \rho_j \leq 1$, is the share of the gatekeeper's rent that provider j captures when access is mandated. The gatekeeper's rent is $\Pi^G = (1 - a) r V^{\text{mobile}}$.

Whether the dissipated rent reaches providers, and how much of it, is an assumption embodied in ρ_j ; with $\sum_j \rho_j < 1$ part of it is competed away to users.

Differentiating gives the three predictions the empirical work tests.

Proposition 3 (Quantity channel). $\partial \Pi_i^B / \partial L = -s \sigma [1 - F_i(L)] \leq 0$, strictly negative whenever $F_i(L) < 1$, and $\partial \Pi_i^B / \partial \lambda = -s C_i \leq 0$, strictly negative when $C_i > 0$. Events that lower the limit or bar legal persons raise bank rent, and the effect is proportional to $1 - F_i(L)$.

Proposition 4 (Fee channel). $\partial \Pi_j^P / \partial \bar{f} = V_j \cdot \mathbf{1}\{\bar{f} < m\} \geq 0$. A tighter cap lowers provider rent in proportion to euro area volume, and leaves bank deposit rent untouched except through ϕ_i .

Proposition 5 (Access channel). $\partial \Pi^G / \partial a = -r V^{\text{mobile}} < 0$ and $\partial \Pi_j^P / \partial a = \rho_j r V^{\text{mobile}} \geq 0$. The access mandate lowers gatekeeper rent unambiguously; it raises provider rent to the extent that $\rho_j > 0$; and it has no first-order effect on bank deposit rent.

4.4 Why the signs can differ within one act

Propositions 3 to 5 operate through different arguments of different rent functions. An act that lowers L and tightens $\bar{f}$ raises Π^B and lowers Π^P simultaneously. The committee position of June 2026 has this form: it limits quantities and caps fees. The Council position of December 2025 is more complex. For banks it combines a political ceiling on holdings, which is favourable, with free basic services and capped compensation, which are costly, and we code its net bank direction as zero. For payment providers it combines a fee cap, which lowers their rent, with mandated device access, which raises it when $\rho_j > 0$; the model does not sign the net effect without an assumption on the relative size of the two channels. We code it as unfavourable, on the reading that the fee cap applies to all providers while access benefits only those that use it, and we report the provider estimate with this event recoded to zero. The empirical design exploits acts with opposite signs across groups, because a confound would have to be correlated with bank exposure and provider exposure with opposite signs on the same day.

4.5 What the model does not deliver

Three limitations are worth stating before the empirical work.

First, the model is static and says nothing about the dynamics of substitution, which is where the run literature operates (Bidder et al. 2024). Our events are one-day announcements about a product that does not yet exist, so the object we measure is the market's present value of a rent transfer, not a run.

Second, the model treats the deposit spread s as fixed. Chiu et al. (2023) show that with deposit market power the spread itself responds, and that the response can be large enough to reverse the sign of the effect on intermediation. Our design cannot separate a small quantity effect from a large quantity effect offset by a spread response; we test the market-power prediction only through cross-sectional heterogeneity, and we say so.

Third, take-up σ is exogenous here. In practice take-up depends on the very design parameters under negotiation, in particular on whether an online version exists at all. The November 2025 draft report, which makes the online version conditional, is best read as an event about σ rather than about L , and we code it accordingly.

5 Data

5.1 Events

We assemble 26 events between June 2023 and September 2026 from four primary sources: the European Parliament’s Legislative Observatory for the procedure file, Council press releases and meeting pages, ECB publications, and the Eurogroup record. Twenty-four are verified directly at the source; the two that are not are excluded from every specification, and one verified event falls after the end of the factor sample. Table 2 lists the main events, Table 8 all of them, and Table 3 gives the sample flow.

5.2 Coding of design content

Each event is coded on five dimensions before any price data is loaded: whether it is purely procedural; the direction of its design content for banks, for payment providers and for device gatekeepers, on a scale of -1 , 0 , $+1$; the quantitative dose where the act names one; and the degree of surprise. The codebook, with the decision rule for each dimension and the documented justification for every event, is in Appendix A. The coding is hash-frozen.

Two features of the coding deserve emphasis. First, the direction is coded *separately by group*, because the acts themselves are multidimensional. Second, the lobbying record is used only to measure surprise and never to determine direction, so that the treatment and the intensity measure are kept apart.

5.3 The negotiation record

The procedure file publishes meetings between members acting in a reporting capacity and interest representatives. We transcribe all 309 meetings between 11 July 2023 and 13 July 2026. Each record has a date, a member, the member’s role and the organisation.

The record is informative in three ways. The time profile tracks the file: 48 meetings in 2023, 20 in 2024 when the dossier lay dormant between parliamentary terms, 161 in 2025, and 80 in the first seven months of 2026. The composition shows that the contest was not one-sided: alongside banks and banking associations, the record contains card networks, European payment schemes, big technology firms, retailers and consumer organisations. And the intensity in the thirty days before an event varies from one meeting, before the procedural event of November 2024, to twenty-two, before the Eurogroup agreement of September 2025.

Limitation. Publication is mandatory only for rapporteurs, shadow rapporteurs and committee chairs. Meetings by other members are published voluntarily and incompletely; we transcribe only the mandatory entries.

5.4 Firms and exposure

The sample contains 51 listed firms in five groups: 30 euro area banks from 11 countries, 10 payment providers and card networks, 2 device gatekeepers, 2 large technology firms on the acceptance side,

Table 2: Events and coded design direction

Date	Event	Banks	Providers	Devices	Source
28.06.2023	Commission proposal COM(2023)0369	—	—		EUR-Lex
19.10.2023	Committee referral announced	0	0		OEIL
31.10.2023	ECB opinion CON/2023/0034	—	0		ECB
09.02.2024	ECON draft report (Berger), never voted	0	0	0	OEIL
20.02.2024	LIBE opinion (privacy)	0	0		OEIL
21.02.2024	Amendments to that draft	0	0	0	OEIL
13.11.2024	Resumption of business	0	0		OEIL
16.12.2024	Change of rapporteur	0	0	0	OEIL
02.09.2025	Rapporteur’s paper published	+	+	0	Yearbook
19.09.2025	Eurogroup: ceiling mechanism	0	0	0	Council
23.10.2025	European Council calls for speed	—	—	0	Council
30.10.2025	ECB closes preparation phase	—	—		ECB
03.11.2025	Draft report: online conditional	+	+	0	OEIL
19.12.2025	Council position	0	—	—	Council
05.03.2026	ECB call for pilot providers	—	0	0	ECB
23.06.2026	Committee position adopted	+	—		OEIL
09.07.2026	Plenary confirms mandate	+	—		OEIL
14.07.2026	ECB selects 36 providers	—	0	0	ECB

Notes: Excerpt; the full list of 26 events with justifications is in Appendix A. Direction is the coded design content for the group, from the contemporaneously available text of the act, assigned before any price data was loaded; the two February 2024 events, whose text was not available, are the exception documented in Appendix A. Blank cells denote acts with no provision affecting that group. Events coded 0 on all three dimensions and marked procedural serve as falsification tests.

and 7 non-euro banks with comparable deposit structures that serve as a control group. Exposure is measured before June 2023 in every case, so that it cannot respond to the events.

For banks, exposure is the share of household and non-financial corporate deposits in total liabilities, from the EBA transparency exercise, and in a second specification the share of overnight deposits of the same sectors. For providers, the planned measure was the share of revenue earned in euro area transactions, from segment reporting; it was not implemented, and providers enter with a group indicator. For gatekeepers, the device-access provisions apply to firms controlling a mobile operating system, so exposure is an indicator rather than a continuous measure. Bank exposure is taken from the EBA transparency exercise for 31 March 2023 (Section 9.6).

5.5 Factor data and two calendar checks

The European three-factor daily series from the Fama and French data library covers our whole sample, from June 2019 to 31 July 2026; over the legislative period, from June 2023 to July 2026, its market factor has an annualised volatility of 13.8%. Two checks on the event calendar follow from it and are worth recording, because both would be easy to get wrong.

Every coded event falls on a trading day. All 23 events with design content or procedural status between June 2023 and July 2026 are trading days in the European calendar. No event needs to be shifted to the next open day, which removes a common source of discretion in event studies of legislative acts, since committee votes and Council meetings are scheduled on working days.

One event lies beyond the factor sample. The ECB’s call for merchants of 15 September 2026 falls after the last factor observation and is excluded. This was anticipated: the event was recorded for an out-of-sample extension rather than for the main specification.

Implied sample size. With 44 firms in the three treated groups and 22 usable events, the planned second stage has 968 firm-event observations before any exclusion for missing prices; with the non-euro control banks and after the exclusions of Section 9, the estimation sample has 1,042. The empirical standard deviation of one-day abnormal returns is 1.6%, between the two noise levels assumed in the power calculation of Section 8.

5.6 Two kinds of event, and the sample flow

The events differ in what they tell the market, and the distinction matters for what our estimate means. *Design-parameter events* fix or signal a parameter of the instrument: the rapporteur’s paper of September 2025, the Eurogroup agreement on the ceiling mechanism, the draft report of November 2025, the Council position of December 2025, and the committee position and its confirmation in plenary in June and July 2026. *Implementation events* change the probability or the timing of issuance without changing its design: the Commission proposal, the ECB opinion, the call of the European Council for speed, the ECB’s move to the next phase, and the call for and selection of pilot participants. The title question concerns the first kind. The pre-specified specification pools both; Section 9.8 separates them.

Table 3: Sample flow of events

Events collected, June 2023 to September 2026	26
less: after the end of the factor sample (15.09.2026)	–1
less: date or content not verified at a primary source	–2
less: merged by the collision rule (two acts on 19.12.2025)	–1
Estimation dates, pre-specified specification	22
less: coding relied on ex-post information (Feb 2024 draft and amendments)	–2
Estimation dates, specification without look-ahead	20
<i>Of the 20, events with a non-zero coded direction for banks</i>	
Design-parameter events	4
Implementation events	6

Notes: Section 8 reports the power calculation on the 22 pre-specified estimation dates; a first run before estimation, on the 19 events coded at that time, reached the same conclusion.

Timing within the day, and same-day confounders. Every event falls on a trading day, but a release after the close of European markets moves prices on the following day and biases a one-day window towards zero. Intraday timing is documented for committee and plenary votes and for some other releases, but not for all. Two events need attention. The European Council conclusions of 23 October 2025, which contain the call for speed on the digital euro, were adopted at the end of a summit day, and the same morning the Council adopted its nineteenth package of sanctions against Russia, which is news for European banks with Russian exposure. The Council position of 19 December 2025 was published on the second day of a European Council summit on the multiannual budget, enlargement and financing for Ukraine. We therefore report two timing checks, fixed from the documented release

circumstances before they were estimated. Moving the European Council event to the next trading day, 24 October 2025, reduces the main estimate from 0.051 to 0.011 percentage points ($t = 0.20$). Excluding both confounded days gives -0.005 ($t = -0.09$). Both checks move the estimate towards zero, which indicates that the small positive coefficient of the pre-specified specification is, if anything, inflated by news unrelated to the digital euro. The two-day window, which absorbs any release after the close, gives 0.13 with $p = 0.15$. We then classify all 22 estimation dates by their documented release circumstances: eight took place during the trading day (committee and plenary votes, the Commission’s midday presentation), one after the close (the European Council conclusions), and for thirteen the release time is not documented. In a timestamp-aligned specification we use the event day for the first group, the following day for the second and both days for the third. This raises the main estimate to 0.16 percentage points per standard deviation (standard error 0.089, 95% confidence interval -0.01 to 0.33), with a permutation p -value of 0.075 and a wild cluster bootstrap p -value of 0.053, and to 0.155 ($t = 1.94$) when the two days with major unrelated news are also excluded. We report this openly because it cuts against the reading that every check pushes the estimate towards zero: the point estimate depends on how the window treats releases of unknown timing, and with the most conservative treatment a small positive effect is significant at the 10% level under both procedures but not at the 5% level. Under a fully euro-denominated market model in excess returns the same alignment gives 0.167, with a wild cluster bootstrap p -value of 0.024 and a permutation p -value of 0.050 (Section 9.8). What does not change is the bound. The upper end of the confidence interval in the timestamp-aligned specification is about 0.33 percentage points, still a third of the magnitude documented for 2020 and 2021.

5.7 Measuring exposure

Exposure is the object that turns a common announcement into a cross-sectional experiment, so we define it carefully and fix it before June 2023 in every case.

Banks. The primary measure is the share of deposits from households and non-financial corporations in total liabilities, taken from the EBA transparency exercise for the reference date closest to, and before, the Commission proposal. This is the measure used by Burlon et al. (2024) and we adopt it so that magnitudes are comparable.

Proposition 1 implies that the marginal effect of changing the limit depends on the fraction of depositors whose balances exceed it. That fraction cannot be measured without account-level distributions, which are not public for the banks in our sample. We therefore use, as a secondary measure, the share of overnight deposits of households and non-financial corporations. This is a proxy for the *substitutable deposit base*, the balances a retail instrument with payment functionality would compete for, and not a direct measure of the marginal effect of the limit; we are explicit that it serves the first implication of Proposition 2 rather than the second.

A third planned measure would address the mechanism in Chiu et al. (2023): the deposit market concentration of the bank’s home market, from ECB structural statistics. Under the market-power channel the valuation response to a given substitution threat should be weaker, or even reversed, where banks have been able to keep deposit rates below the market rate. The concentration data were not

assembled, so this measure is not used (Appendix E).

Payment providers. The planned exposure measure is the share of revenue earned on euro area transactions, from segment reporting in the annual accounts for the last financial year before the proposal. We did not assemble these data, so the provider coefficient is estimated with a group indicator throughout, a deviation from the plan recorded in Appendix E. The remainder of this paragraph describes what the planned measure would capture. For the international card networks this share is modest in relation to global revenue, which predicts a smaller proportional valuation response and is itself a testable implication rather than a nuisance.

We planned to split providers by role, following the distinction the ECB itself uses in the pilot: distributing providers, which face consumers, and acquiring providers, which connect merchants. The fee cap binds on the acquiring side, so the fee channel should load on acquirers. The split requires provider-level role and revenue data that we did not assemble and is not implemented.

Device gatekeepers. The access provisions apply to firms that control a mobile operating system. Exposure is therefore an indicator, and the group contains two firms. We do not attempt a continuous measure and we do not claim to identify a dose for this channel; the prediction is a sign on the access event.

What could go wrong with exposure. Two risks deserve statement. First, the deposit share is correlated with other bank characteristics, notably size and business model, which could respond differently to unrelated news on the same day. The firm fixed effect absorbs the level, and only the alternation of the coded sign identifies the coefficient, but a characteristic whose return loading switches sign with our coding would not be absorbed. We know of no such characteristic and we say so rather than claiming there is none.

Second, exposure is measured with error, and classical measurement error in an interacted regressor attenuates the coefficient towards zero. A null would therefore be weaker evidence than a positive finding of the same precision, and we read the results with this in mind.

6 Empirical strategy

6.1 Two stages

In the first stage, abnormal returns are computed from a three-factor model (Fama and French 1993),

$$R_{i,t} = \alpha_i + \beta_i^m R_t^m + \beta_i^{smb} R_t^{smb} + \beta_i^{hml} R_t^{hml} + \varepsilon_{i,t}, \quad (2)$$

estimated once per firm on the most recent 250 trading days that lie at least ten days away from any event. This pre-specified common window uses observations after the event for the early events; we therefore report the conventional strictly pre-event estimator, with loadings estimated separately for each event on the 250 days before it, as an important robustness check (Table 10). Cumulative abnormal returns are taken over a one-day window, with a two-day window as a robustness check. For firms listed in the United States the first stage is a market model on the S&P 500 (Section 9).

In the second stage,

$$CAR_{i,e} = \alpha_i + \delta_e + \theta (\text{Exposure}_i \times \text{Direction}_{g(i),e}) + u_{i,e}, \quad (3)$$

where $\text{Direction}_{g,e}$ is the coded content of event e for the group g to which firm i belongs. Event fixed effects δ_e absorb everything common to a trading day, including market-wide news and the average effect of the event itself. Firm fixed effects absorb persistent differences in abnormal returns. In the continuous specification of Section 9.6 the regression also contains the group-level direction terms, so that θ is identified from differences in exposure within the bank group.

6.2 What identifies θ

Two sources of variation. Within an event, firms differ in exposure, so the interaction varies across the cross-section. Across events, the coded direction changes sign, so a firm with high exposure should gain on limiting events and lose on expanding ones. The second source is what distinguishes this design from a study of project news: a level effect common to all events is absorbed by the firm fixed effect, and only the alternation of signs identifies θ .

6.3 Inference

Treatment is coded at the event level and exposure varies at the firm level, so the effective number of independent units is small. We report three procedures. Regression t -statistics use leave-one-firm-out jackknife standard errors. A permutation test reassigns exposure values across firms *within group*, holding the event coding fixed, and tests the null that exposure carries no information. We also report a null-imposed wild cluster bootstrap with Rademacher weights, clustered on the firm (Cameron et al. 2008). We assess finite-sample size by simulation before applying the procedures. All permutation and wild bootstrap p -values use 9,999 draws.

6.4 Pre-specified falsification tests

- F1 Procedural events with no design content show no effect. Four estimation dates are coded procedural: the referral of the proposal to committee in October 2023, the resumption of business after the parliamentary term in November 2024, the appointment of a new opinion rapporteur in July 2025 and the Rule 72 announcement in plenary in July 2026.
- F2 (Planned; not identified in the binary specification.) Non-euro control banks with comparable deposit structures show no effect on any event. In the binary specification this test is collinear with the event fixed effects; it is replaced by the placebo-date test, and in the continuous specification it is implemented as the non-euro exposure placebo of Section 9.8.
- F3 The sign reverses between limiting and expanding events, which is tested by splitting the sample rather than by the interaction alone.
- F4 Placebo dates produce no effect. The plan drew them from the same weeks as the events; the implementation draws them from non-event trading days, as recorded in Appendix E.

F5 Device gatekeepers respond only to the event containing the access provision, and not to quantity events.

6.5 Robustness and multiple testing

The specification is fixed in advance, and the following variations are reported in Table 10 in Appendix D rather than selected; one was not implemented and is listed among the deviations.

Event window. The main window is the event day alone. A two-day window is reported because several acts are published late in the trading day; a three-day window is reported as a bound, with the caveat that the December 2025 acts and the July 2026 acts fall close enough together that wider windows overlap.

Estimation window for the factor model. The main estimates use the 250 most recent clean trading days with a ten-day exclusion band around each event. We report 120 and 500 day alternatives, and a specification that excludes the March 2023 banking turmoil from the estimation window for all firms.

Factor model. The three-factor model is the baseline for comparability with Burlon et al. (2024). We report a market-model specification. A planned four-factor specification adding a European banks index orthogonalised to the market could not be implemented because the index series was unavailable; this deviation is recorded in Appendix E.

Exposure measure. Deposit share and overnight deposit share are reported side by side rather than chosen; a concentration measure was planned but not assembled.

Event set. Each event is dropped in turn. The two events that could not be verified at a primary source are excluded from every specification and listed in Table 8. Two events are re-dated as an alternative: the Council position to the date of its working document and the draft report to the date of its presentation in committee.

Treatment of collisions. Collisions are handled by a rule fixed in advance. The Council position and the tabling of committee amendments both fall on 19 December 2025; they are treated as a single event with combined coding. The first trilogue was held the day before the selection of pilot participants in July 2026; its date could not be verified at a primary source, so it is excluded and the selection enters alone.

Multiple testing. We test three group-specific predictions and four heterogeneity splits. The three primary predictions are treated as a family and adjusted with the step-down procedure of Romano and Wolf (2005), using the placebo-date resampling described in Section 9 to approximate the joint null distribution while preserving the common event structure. The heterogeneity splits are reported unadjusted and labelled exploratory.

6.6 Heterogeneity and mechanisms

Beyond the average effect, four splits are specified in advance because each distinguishes between competing accounts of the same coefficient.

Market power. Under the account in Chiu et al. (2023), the threat of substitution disciplines banks that have been exercising deposit market power, so the valuation loss from a higher limit should be

concentrated where concentration is high, while the effect on lending could have the opposite sign. The planned split was at the median of home-market deposit concentration; it is not implemented because the concentration data were not assembled. A coefficient that is uniform across the split is evidence for the mechanical substitution account; one that loads on concentrated markets is evidence for the competition account.

Balance composition. Proposition 1 says the marginal effect of the limit is proportional to the fraction of depositors with balances above it. Splitting banks by that fraction would test the shape of the channel rather than its existence; without account-level data we approximate it with the overnight share, which is a weaker test, and is the sharpest available test that the deposit channel, rather than a generic bank factor, is what we are picking up.

Role within the payments group. The fee cap binds on the acquiring side. If the provider coefficient is driven by acquirers rather than by distributing providers, the fee channel is confirmed; if it is uniform, the estimate is more likely to reflect a general sentiment towards the payments sector.

Home country. The file is negotiated by national delegations, and the negotiation record shows a marked concentration of meetings with Spanish, German and French institutions. If the coefficient loads on banks from countries whose representatives held the pen, the reading shifts from pure economics towards the political economy of the file. We address this dimension through a country-blocked permutation and country-residualised exposure, and treat it as descriptive, since home country correlates with much else.

Which of these were implemented. The balance-composition split is implemented through the overnight deposit share (Table 6) and the home-country dimension through the country-blocked permutation and the country-residualised exposure (Section 9.6). The market-power split and the provider-role split require concentration data and provider-level revenue data that we did not assemble; they are recorded as deviations from the plan.

6.7 Threats

Anticipation. Legislative steps are scheduled and partly foreseeable. We address this in three ways: by coding surprise before estimation, by using the negotiation record as an ex-ante measure of how contested an event was, and by restricting the main specification to a one-day window. We do not claim that the events are unanticipated; we claim that the residual surprise is systematically related to the coded content.

Confounding news. Events fall on days with other financial news. Event fixed effects absorb anything common, and the cross-sectional interaction is identified from differences between firms on the same day. A confound would have to be correlated with exposure and to switch sign with the coded direction.

Group composition. Banks, providers and gatekeepers differ in many ways besides their exposure to the digital euro. The within-group variation in exposure and the non-euro control group address this; the selection of pilot participants would provide a within-group comparison, but a usable firm-level list of the selected providers was not available, and we record this as an open item rather than a result.

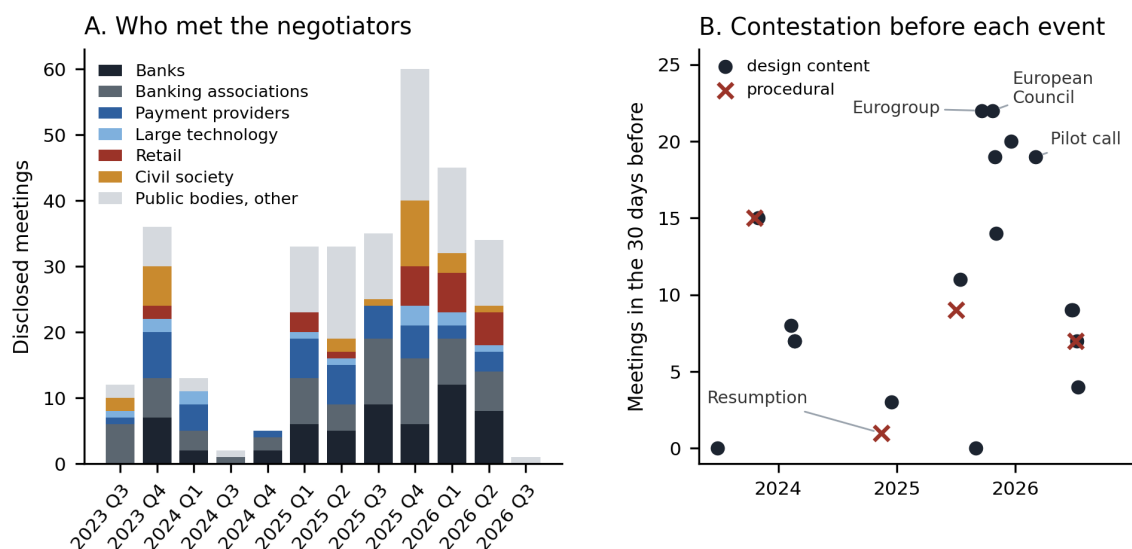


Figure 2: The negotiation record for the digital euro file, 2023–2026

Notes: All 309 meetings disclosed on procedure file 2023/0212(COD) by rapporteurs, shadow rapporteurs and the committee chair, July 2023 to July 2026. Categories assigned by the author; public bodies are mostly central banks, finance ministries and permanent representations. Panel B counts meetings in the 30 calendar days before each coded event.

7 The negotiation record

The procedure file publishes every meeting between a member acting in a reporting capacity and an interest representative. We transcribe the complete published record for the digital euro file: 309 disclosed meetings between 11 July 2023 and 13 July 2026, of which 145 were held by the rapporteur, 161 by shadow rapporteurs and the remaining three by the committee chair and by shadow rapporteurs for the opinion of the civil liberties committee. This section describes what the record shows and how we use it.

Who bargained. Figure 2, panel A, shows the composition by quarter. Banks and banking associations account for 119 of the 309 meetings, payment providers and schemes for 40, retailers for 23, consumer and civil society organisations for 25, and large technology firms for 13. The residual category, 89 meetings, consists mostly of central banks, finance ministries and permanent representations, which we exclude from any measure of private interest.

The most frequent single counterparties are Deutsche Bank with twelve meetings, the French banking federation with ten, Amazon and the European consumer organisation with eight each, and Banco Santander and the European retail association with seven each. The presence of card networks, European payment schemes, retailers and device manufacturers alongside banks is the first piece of evidence that the incidence of this file is not confined to the deposit channel, and it is evidence that does not come from prices.

When they bargained. The time profile tracks the legislative calendar rather than the news cycle: 48 meetings in 2023 around the Commission proposal, only 20 in 2024 when the file lay dormant between parliamentary terms, 161 in 2025 as the Council and the new rapporteur converged on positions, and 80 in the first seven months of 2026 up to the plenary mandate.

How we use it. Panel B plots the number of meetings in the thirty days before each coded event, separating events with design content from procedural ones. Two features are relevant for the empirical design.

First, the peaks fall where the model says the stakes are: 22 meetings before the Eurogroup agreement on the ceiling mechanism of September 2025, of which 14 were held with banks or banking associations, and 19 before the ECB's call for pilot participants in March 2026, of which 13 were with the banking side.

Second, the procedural event of November 2024, the formal resumption of business after the parliamentary term, is preceded by a single meeting, against a mean of 11 before events with design content. This matters because it validates our falsification test from outside the price data. We coded that event as procedural from its content; the people who had money at stake behaved as if it were.

What we do not claim. The record is used descriptively and as a measure of how contested an event was. We do not use it to determine the coded direction of any event, and we do not treat it as exogenous: firms lobby because they expect to be affected, so lobbying intensity and exposure are jointly determined. Using intensity as an instrument would be indefensible, and we do not. Publication is also mandatory only for members in a reporting capacity, so the record is a lower bound on contact rather than a census of influence.

8 Power

Event studies with a handful of events and a few dozen firms can be badly underpowered, and the discipline of saying so before estimation is part of the design. We simulate the second stage using the actual structure of the data: the coded direction of each event by group, the planned firm universe and the number of firms in each group. Only the noise in abnormal returns and the distribution of exposure are assumed.

Table 4 reports the results. With 44 firms in the three treated groups and the 22 pre-specified estimation dates, of which ten carry a non-zero direction for banks and eight for providers, and with a standard deviation of daily abnormal returns of 1.5%, the design detects an effect of 0.75 percentage points per unit of the exposure-direction interaction with probability 0.86. At the empirical standard deviation of 1.6% the probability is 0.82. Under the more conservative assumption of 2.5%, power of 0.86 is reached at 1.25 percentage points. Simulated size lies between 0.05 and 0.06. We first ran the simulation before estimation, on the 19 events coded at that time, and reached the same conclusion.

Is this enough? The natural benchmark is the published estimate for this market. Burlon et al. (2024) report that a one standard deviation difference in the deposit ratio, 18 percentage points, is associated with more than one percentage point of abnormal return per event. In the units of our interaction that corresponds to roughly 5.5 percentage points, more than four times our minimum detectable effect even under the conservative noise assumption. The design is therefore well powered against effects of the size already documented in this market, and would detect effects several times smaller. It is not powered against effects an order of magnitude smaller than the published benchmark, and we report bounded nulls as such rather than as evidence of no effect.

Table 4: Power of the second stage

Noise in abnormal returns	Effect θ in percentage points per unit of interaction					
	0	0.50	0.75	1.00	1.25	1.75
$\sigma_{\text{CAR}} = 1.5\%$	0.06	0.56	0.86	0.98	1.00	1.00
$\sigma_{\text{CAR}} = 1.6\%$ (empirical)	0.05	0.50	0.82	0.97	1.00	1.00
$\sigma_{\text{CAR}} = 2.5\%$	0.06	0.25	0.46	0.71	0.86	0.99

Notes: Rejection rate of a two-sided test at the 5% level with leave-one-firm-out jackknife t -statistics, 2,000 replications per cell. The simulation uses the coded direction of each of the 22 pre-specified estimation dates by group and the planned firm universe (44 firms in the bank, provider and gatekeeper groups); it uses no price data. The row at 1.6% uses the empirical standard deviation of one-day abnormal returns. Exposure is drawn uniformly on $[0.2, 0.8]$, so an interquartile move in exposure is about 0.3 units and the implied effect on abnormal returns is roughly 0.3θ . The column at $\theta = 0$ reports size.

9 Results

We first report the group-level specification, fixed in advance for the case in which bank-level deposit data were not available. In that specification exposure is an indicator for membership of the treated group, which is the binary special case of equation (3). The continuous specification with deposit shares is the pre-specified main test and uses variation within the bank group; Section 9.6 reports it. The event coding and the firm universe were hash-frozen before any return was computed.

9.1 Sample

Price data were obtained for 51 of 54 series. Three failed to download: Bank of Cyprus, Fiserv and the STOXX Europe banks index. Alpha Services is available only from July 2025, after a change of listing, and drops out of earlier events for want of an estimation window. The estimation sample has 49 firms, 22 events and 1,042 firm-event observations in the one-day window. Euro area firms are benchmarked against the European three-factor model as pre-specified; firms listed in the United States are benchmarked against a market model on the S&P 500, a choice logged before any abnormal return was computed, because European factors are not a meaningful benchmark for US equities.

9.2 Main estimates

Table 5 reports the estimates. None is statistically distinguishable from zero at conventional levels.

Banks. The bank coefficient is 0.10 percentage points per unit of coded direction in the one-day window and 0.18 in the two-day window. It has the predicted sign. It is not distinguishable from random trading days: moving the actual direction vector to 500 sets of non-event days produces a distribution with a 95% band from -0.50 to 0.54 percentage points, and the actual coefficient sits well inside it, with a placebo p -value of 0.62 (Figure 3, panel B). The placebo exercise moves only the bank direction vector, so it uses the regression with the bank term alone, whose coefficient is 0.118 rather than the 0.101 of the full specification with all group terms; the two differ only in whether the provider and gatekeeper terms are included. The placebo band gives the most useful statement of what the design can exclude: an average differential response of euro area banks larger than about

Table 5: Effect of coded design direction on abnormal returns, group-level specification

	One-day window		Two-day window	
	Estimate	<i>t</i>	Estimate	<i>t</i>
Pooled across groups	0.055	0.37	0.093	0.47
Banks	0.101	0.62	0.186	0.82
Payment providers	−0.147	−0.36	−0.287	−0.94
Device gatekeepers and large technology	0.867	1.68	1.400	1.60
Firms / events / observations	49 / 22 / 1,042		49 / 22 / 1,032	
Placebo-date <i>p</i> -value, banks	0.62			

Notes: Estimates are in percentage points of abnormal return per unit of the coded design direction, which takes the values -1 , 0 and $+1$; under the hypothesis each coefficient is positive. Exposure is an indicator for membership of the treated group; non-euro banks are the reference group with direction zero on every event. All specifications include firm and event fixed effects. *t*-statistics use leave-one-firm-out jackknife standard errors. The placebo-date *p*-value compares the bank coefficient from a regression with the bank term alone (0.118) with its distribution when the actual coded direction vector is moved to 500 sets of random non-event trading days.

half a percentage point per unit of coded content.

Payment providers. There is no consistent pattern. The one-day and two-day coefficients are both small and negative, the opposite of the prediction, and neither is distinguishable from zero. The Council position of December 2025 combines a fee cap with mandated device access, whose net effect on providers the model does not sign; recoding its provider direction to zero gives -0.38 ($t = -1.16$), again not distinguishable from zero. The group combines international card networks, euro area acquirers and a digital wallet, and it contains two firms that suffered very large idiosyncratic price falls during the sample for reasons unrelated to the file. We do not read the two-day estimate as evidence against the fee channel; we read the group as too heterogeneous for the binary specification to be informative.

Device gatekeepers. The coefficient is the largest and has the predicted sign, but it cannot bear weight. The group has four firms, and only one event carries a non-zero direction for it, the Council position of 19 December 2025, which contained the device-access provision. Dropping that single event sets the coefficient to zero. This is one observation per firm, and we report it as such.

9.3 Event by event

Figure 3, panel A, shows what lies behind the bank coefficient. For each of the ten events with a non-zero coded direction for banks, it plots the mean one-day abnormal return of euro area banks minus that of non-euro banks. The gap has the predicted sign in eight of the ten events. This observation was not part of the pre-specified analysis, and we report it as exploratory: a one-sided sign test gives $p \approx 0.055$, which is suggestive and no more. The regression coefficient is smaller than the average signed gap because the firm fixed effects absorb the fact that euro area banks earned positive abnormal returns throughout the 2023 to 2026 rally, including on days without design content.

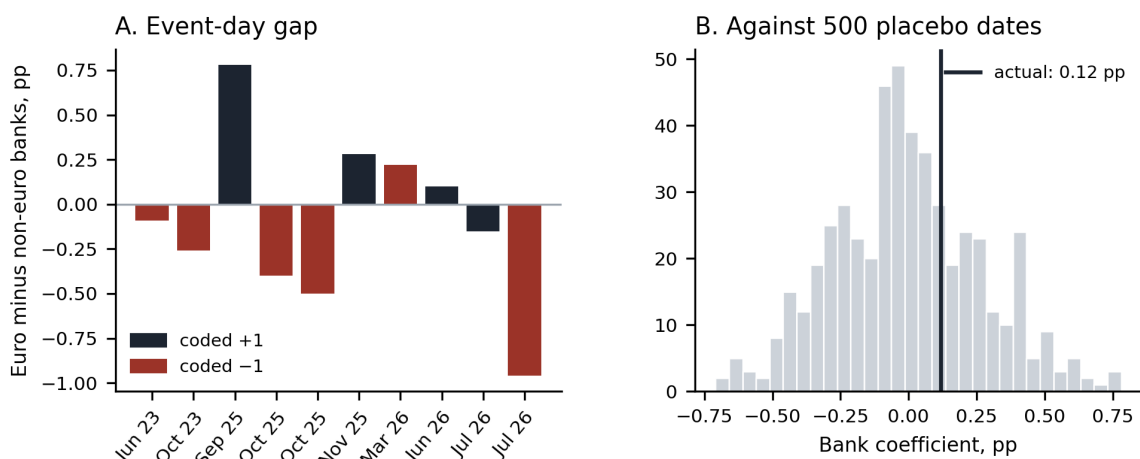


Figure 3: Banks: event-day abnormal return gap and placebo distribution

Notes: Panel A: for the ten events with a non-zero coded direction for banks, the mean one-day abnormal return of 29 euro area banks minus that of 7 non-euro banks. Panel B: the bank coefficient from a regression with the bank term alone (0.118; firm and event fixed effects) against its distribution when the coded direction vector is moved to 500 sets of random non-event trading days. The full specification with all group terms gives 0.101.

9.4 A falsification test we discarded, and why

The analysis plan specified a test in which non-euro banks are assigned the same coded direction as euro area banks. In the group-level specification that test is not identified: the two regressors sum to a function of the event alone and are absorbed by the event fixed effects, so only their difference is estimable, and that difference is the main estimate itself. We computed it before recognising this, obtained coefficients that are artefacts of the collinearity, and discarded them. We replaced the test with the placebo-date randomisation, which is valid in the binary design. We report the episode because the plan promised the original test.

9.5 What these results do and do not say

They say that, measured at the level of the treated group, the legislative design content of the digital euro did not move the relative valuations of euro area banks, payment providers or device gatekeepers by more than about half a percentage point per event. They do not speak to the channel within banks that the model identifies, namely that banks with a larger substitutable deposit base should respond more. That test requires bank-level deposit composition and is the pre-specified main specification. It is also the more powerful one: the binary specification uses only the difference between euro area and non-euro banks, and discards the variation in exposure across 29 euro area banks.

9.6 Continuous exposure: the pre-specified main specification

The pre-specified test interacts the coded direction with each euro area bank's deposit exposure, measured on 31 March 2023, the last reporting date before the Commission proposal, from the EBA transparency exercise. The primary measure is deposits of households and non-financial corporations over total liabilities, as in Burlon et al. (2024). The secondary measure, derived from Proposition 1,

Table 6: Deposit exposure and abnormal returns: pre-specified main specification

	Estimate	Std. error	<i>t</i>	Permutation <i>p</i>
<i>One-day window</i>				
Deposit share (main)	0.051	0.056	0.92	0.45
Overnight deposit share	0.060	0.054	1.12	0.35
Deposit share, excluding Crédit Agricole	0.053		0.94	
<i>Two-day window</i>				
Deposit share	0.131	0.092	1.42	0.15
Overnight deposit share	0.075	0.084	0.90	0.41

Notes: Estimates are percentage points of abnormal return per standard deviation of exposure per unit of coded design direction; under the hypothesis they are positive. Exposure is measured on 31 March 2023 and standardised across the 29 euro area banks. The regression also contains the group-level direction terms of Table 5 and firm and event fixed effects. Standard errors are leave-one-firm-out jackknife. The permutation *p*-value reassigns exposure across euro area banks, 9,999 draws. Crédit Agricole is excluded in one row because the supervisory data refer to the consolidated group while the listed entity is Crédit Agricole SA.

uses only overnight deposits of the same sectors, since these are the balances a holding limit directly captures. Both measures and the specification were fixed and hash-frozen before estimation. Across the 29 euro area banks the deposit share averages 59% with a standard deviation of 14 percentage points, ranging from 32% at Mediobanca and Société Générale to 87% at Banco Comercial Português; the two measures correlate at 0.87. Measures are standardised within euro area banks.

Table 6 reports the results. Every coefficient has the predicted sign, and none is statistically distinguishable from zero. In the main specification a bank one standard deviation more exposed to deposit substitution gains 0.05 percentage points more on an event coded as limiting the digital euro, with a permutation *p*-value of 0.45. The overnight measure, our mechanism-motivated proxy for the substitutable deposit base, gives 0.06. The two-day window is larger at 0.13 with *p* = 0.15, still inconclusive. Excluding the bank whose supervisory and listed perimeters differ changes nothing.

What the exposure permutation assumes. Deposit funding is not randomly assigned across banks; it varies with country, business model and size. Permuting exposure across euro area banks is therefore not design-based randomisation inference. It is valid under exchangeability of the exposure labels conditional on the fixed effects: if exposure has no effect on event-day returns, reassigning it across banks leaves the distribution of the statistic unchanged. Two variants make the assumption weaker. Permuting only within country, so that a German bank's exposure is only ever assigned to another German bank, gives *p* = 0.43 across ten country blocks. Residualising exposure on country, so that only within-country differences in deposit funding identify the coefficient, gives 0.011 percentage points per standard deviation (*t* = 0.23). Neither changes the conclusion.

Robustness of the main estimate. Several checks bear on the reading of Table 6. The pre-specified variations of window, estimation sample, factor model and event dating in Table 10 leave the estimate between −0.014 and 0.061; a conventional per-event pre-event estimation window gives 0.099 (*t* = 1.83), in the same range as the release-time-aligned estimate discussed in Section 5.6. Dropping each event in turn moves the main coefficient between −0.005 and 0.153 percentage points; the upper value arises when the ECB's call for pilot participants of 5 March 2026 is removed, an event

on which the most deposit-funded banks outperformed the least deposit-funded ones against the coded direction. A null-imposed wild cluster bootstrap with Rademacher weights on firms gives $p = 0.33$. Restricting the sample to the five events coded as highly surprising before any price was seen gives 0.014 ($t = 0.11$). The last check matters for interpretation: if the null reflected anticipation of scheduled steps, the unscheduled and contested events should show the effect, and they do not. The pre-specified falsification test F1 also holds: adding the interaction of exposure with an indicator for the four procedural dates to the main specification gives a slope of 0.04 percentage points per standard deviation on those dates ($t = 0.55$, permutation $p = 0.73$) and leaves the main estimate at 0.05.

Event by event. Figure 4 in Section 9.7 decomposes the estimate. For each bank-relevant event it shows the cross-sectional slope of abnormal returns on deposit exposure, signed so that positive values agree with the prediction. Six of the ten slopes are positive, and four of them are individually around 0.4 percentage points per standard deviation: the ECB opinion of October 2023, the rapporteur’s paper of September 2025, the European Council of October 2025 and the draft report of November 2025. The events of 2026 are zero or negative, and the pilot call of March 2026 is strongly negative. This pattern is exploratory, since the pre-specified analysis pools the events, and with ten events it can be noise. It is nonetheless informative about the size of the effects at stake. Where the slopes agree with the prediction, they are around 40% of the magnitude reported for 2020 and 2021; nowhere do they approach it; and the events at which the legislative outcome was confirmed, in June and July 2026, show nothing.

Multiple testing and the provider exposure measure. The plan specified two things for the non-bank groups that we report here in the form the data allow. First, a continuous exposure measure for payment providers, their share of revenue earned on euro area transactions. We did not assemble segment-level revenue data for these firms, so the provider coefficient is estimated with a group indicator throughout; this is a deviation from the plan and is recorded as such. Second, the three group-level predictions were to be adjusted jointly with the step-down procedure of Romano and Wolf (2005). Using the placebo-date resampling of Section 9 to approximate the joint null distribution, the adjusted p -values are 0.88 for banks, 0.88 for payment providers and 0.85 for device gatekeepers, against unadjusted p -values from the same joint resampling of 0.65, 0.73 and 0.40. The unadjusted bank value differs from the 0.62 of Table 5 because the joint resampling studentises the coefficients of the full specification with all three group terms over 300 draws, while Table 5 uses the regression with the bank term alone and 500 draws.

What the null excludes. The 95% confidence interval of the main estimate runs from about -0.06 to $+0.16$ percentage points per standard deviation. Burlon et al. (2024) find that during the 2020 and 2021 announcements a one standard deviation higher deposit ratio was associated with more than one percentage point lower abnormal return per event. Allowing for the difference in the sign convention, their magnitude lies more than six times above the upper end of our interval. The legislative design events of 2023 to 2026 therefore moved bank valuations, per unit of deposit exposure, by at most a small fraction of what the ECB’s early announcements did.

Was the design priced earlier? A natural reading of the null is that the market learned the essential design before the legislators took up the file. Burlon et al. (2024) document that the trend in bank

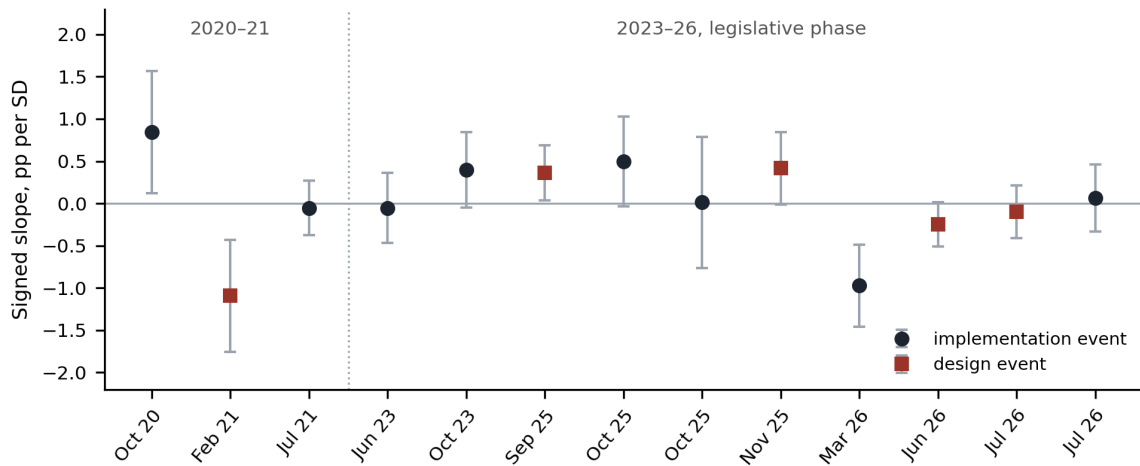


Figure 4: Deposit exposure and abnormal returns, event by event, 2020 to 2026

Notes: For each event with a non-zero coded direction for banks, the cross-sectional slope of the one-day abnormal return of euro area banks on their standardised deposit share, multiplied by the coded direction so that positive values agree with the prediction. Bars are 95% intervals. Deposit shares are measured on 31 December 2019 for the 2020 to 2021 events and on 31 March 2023 for the legislative events.

valuations reversed after the Executive Board first mentioned a holding limit of about EUR 3,000 in February 2021. If the cap was priced then, its later legislative refinement carried little news. We test this directly in Section 9.7.

9.7 Extending the sample to 2020 and 2021

We extend the price data back to June 2019 and code six events of 2020 and 2021 from the ECB’s own record, verified at the relevant press releases: the Eurosystem report on a digital euro of 2 October 2020, the launch and the close of the public consultation, the Executive Board’s statement of 10 February 2021 that a holding limit of around EUR 3,000 was under consideration, the publication of the consultation results, and the launch of the investigation phase on 14 July 2021. Three carry a non-zero direction for banks: the report and the investigation phase, which advance the project (−1), and the holding-limit statement, which is the first quantitative signal of a cap (+1). The consultation steps carry no design content. As before, the coding was fixed and hash-frozen before any return from the new price file was computed, and the test was specified in advance: the same regression as the main specification, on the early events, with the prediction that the early coefficient is large and positive.

The pre-specified test does not support the reading. Exposure for the early events is measured on 31 December 2019, from the EBA transparency exercise of spring 2020, so that it is predetermined; its correlation with the 2023 measure across the 29 euro area banks is 0.97. The early coefficient is −0.27 percentage points per standard deviation of deposit exposure, with $t = -1.47$ and a permutation p -value of 0.14. It has the wrong sign and is not distinguishable from zero. The claim that the cap was priced in 2021 is not supported by our data.

But the design can see a Burlon-sized effect when there is one. Figure 4 shows why the pooled coefficient is uninformative about the early period. On the day the Eurosystem published its report, the most deposit-funded banks underperformed the least deposit-funded ones by 0.85 percentage points per standard deviation of exposure (standard error 0.37), with the predicted sign and a magnitude close to the one reported by Burlon et al. (2024) for the same period. On the day the holding limit was first floated, the slope was -1.09 (standard error 0.34): the most deposit-funded banks again underperformed, which is the opposite of what a favourable design signal predicts. The launch of the investigation phase shows nothing.

Two conclusions follow. First, the report day is a positive control. With the same firms, the same exposure measure, the same factor model and the same estimator, the design detects an effect close to the size documented in the literature when the news is about whether a digital euro will exist. The null for the legislative phase is therefore not a failure of the design to see anything. Second, neither the first cap signal in 2021 nor the legislative design events of 2023 to 2026 moved deposit-funded banks in the direction the design channel predicts. The pattern is consistent with the market having reacted to news about the existence of the project, early, rather than to news about its design; with a single early event of this kind we state it as a pattern, not as an established fact.

Three remarks on robustness. The baseline first stage estimates factor loadings on a common window that lies after these events; estimating them instead on the 250 trading days strictly before each event gives a report-day slope of 0.77 (standard error 0.36) with the European three-factor model and 0.82 (0.37) with a euro-denominated market model in excess returns, and a holding-limit-day slope of -1.09 in both, so the positive control does not depend on post-event information. Using the 2023 exposure measure for the early events instead gives a report-day slope of 1.06, a holding-limit-day slope of -1.21 and a pooled coefficient of -0.24 , so the choice of measurement date does not matter. Finally, single-day slopes can be driven by news unrelated to the digital euro; the report-day estimate is an individual event, not a replicated pattern, and we present it as a demonstration that the design has power rather than as a finding in its own right.

9.8 Additional analyses

The analyses in this section were added after a review of the first complete draft. They were not part of the pre-specified plan and are reported as such, alongside the pre-specified estimate rather than in place of it.

Currency and a correction to the window rule. The pre-specified first stage regresses local-currency returns on the European Fama–French factors, which are computed in US dollars. Two alternatives address this. Replacing only the market factor by the euro-denominated STOXX Europe 600 return, while keeping the dollar-based size and value factors, gives 0.042 in the one-day window, 0.115 in the two-day window, 0.147 in the release-time-aligned specification and -0.003 on design events. A fully euro-denominated market model in excess returns, with the ECB deposit facility rate as the risk-free rate, gives 0.054 in the one-day window ($t = 0.95$) and -0.001 on design events; in the release-time-aligned specification it gives 0.167 (standard error 0.082, 95% confidence interval 0.01 to 0.33), with a permutation p -value of 0.050 and a wild cluster bootstrap p -value of 0.024 (Table 10).

Table 7: Additional analyses of the deposit-exposure specification

	Estimate	<i>t</i>	Permutation <i>p</i>	Events
Pre-specified specification	0.051	0.92	0.45	22
Without events coded with ex-post information	0.051	0.93	0.45	20
<i>Separating the kinds of event</i>				
Design-parameter events	−0.004	−0.04	0.97	4
Implementation events	0.096	0.95		6
<i>Timing</i>				
Timestamp-aligned, all events	0.161	1.82	0.075	22
Timestamp-aligned, excluding confounded days	0.155	1.94		20
Timestamp-aligned, design-parameter events only	0.164	1.07		4
European Council event moved to next trading day	0.011	0.20		22
Excluding the two days with major same-day news	−0.005	−0.09		20
<i>Placebo: the same exposure slope for non-euro banks</i>				
Euro area banks	0.051			
Non-euro banks (four)	0.182	1.10		
Difference, euro minus non-euro	−0.131			

Notes: Percentage points of one-day abnormal return per standard deviation of the deposit share per unit of coded direction. Non-euro banks are assigned the coded bank direction of each event and their deposit share standardised on the euro area scale; the four are Danske Bank, Swedbank, Handelsbanken and DNB; the three non-euro banks without EBA data are excluded from this test. The pre-specified specification is that of Table 6. Standard errors are leave-one-firm-out jackknife; permutation reassigns exposure across euro area banks, 9,999 draws; the null-imposed wild cluster bootstrap *p*-value of the timestamp-aligned specification is 0.053. “Events” counts estimation dates, or for the split, dates with a non-zero bank direction of each kind.

The one-day and design-event results are unchanged by the benchmark. The release-time-aligned estimate is where the benchmark matters: under the cleanest currency treatment it is significant at the 5% level by the bootstrap and on the 5% boundary by permutation. We read this as some evidence of a small effect under conservative timing, and we do not treat it as robust. In the course of these checks we also found that two-day windows had been summed over whichever days had a return rather than requiring both; the corrected rule reduces the two-day sample from 1,069 to 1,032 observations, leaves the bank estimates essentially unchanged, and reduces the two-day payment-provider coefficient from −0.51 to −0.29. All results reported here use the corrected rule.

No look-ahead. Excluding the two 2024 events whose pre-specified coding relied on the later fate of the draft leaves the estimate unchanged at 0.051 percentage points, because both carried a zero direction and entered only through the fixed effects.

Design parameters alone. Restricting the treatment to events that fixed or signalled a design parameter gives an estimate of −0.004 percentage points per standard deviation, with a permutation *p*-value of 0.97. The small positive coefficient of the pooled specification comes entirely from implementation events, and even there it is not distinguishable from zero. This is the sharpest answer to the question in the title, and it should be read with its precision. Four design-parameter events carry a bank direction; the standard error is 0.11 and the 95% confidence interval runs from −0.22 to 0.21 percentage points per standard deviation. We therefore detect no price effect of the design parameters, and the interval excludes effects of the size documented for 2020 and 2021 by a factor of

about five, but it does not exclude small effects. Two further statements make the precision explicit. First, simulating the design-only coefficient on the actual structure of the data, four design events, 29 euro area banks with their observed exposure and the empirical dispersion of abnormal returns, the test has power of 0.54 against an effect of 0.4 percentage points per standard deviation, 0.77 against 0.5 and 0.90 against 0.6, with simulated size of 0.07. The design-only test is therefore informative against effects of about half the documented magnitude and not against smaller ones. Second, the null does not come from a single event. The four event-level slopes, signed so that positive values agree with the prediction, are 0.36 (the rapporteur’s paper), 0.42 (the draft report), -0.24 (the committee position) and -0.10 (the plenary mandate): two in each direction, none large. With four events a sign test carries no information, and we do not report one as evidence. In the timestamp-aligned specification the design-only coefficient is 0.16 (standard error 0.15), again not distinguishable from zero.

Placebo on non-euro banks. If the exposure slope reflected the digital euro, it should be absent for banks outside the euro area, which have comparable deposit structures and no exposure to the instrument. Assigning the same coded directions to the four non-euro banks with supervisory data gives a slope of 0.18 percentage points per standard deviation, larger than for euro area banks, with $t = 1.10$. The difference between euro area and non-euro banks is -0.13 . Whatever small co-movement exists between deposit funding and returns on event days does not appear to be specific to the euro area. It is consistent with a generic factor loading on deposit-funded banks rather than a response to the digital euro, although four control banks cannot establish this. With four control banks the test is weak, but it points in the direction that reinforces the null rather than the hypothesis.

10 Conclusion

A central bank digital currency redistributes rents along the payments chain, and the size of that redistribution is set by its design parameters. We measured the price of those parameters over the three years in which the European co-legislators adopted and narrowed their positions on them, coding each act by its content for banks, payment providers and device gatekeepers before looking at any price, and complementing the price evidence with the complete record of who negotiated with the legislators.

The answer is that the legislative phase moved valuations little, and at most modestly. In the pre-specified specification, a euro area bank one standard deviation more exposed to deposit substitution gained 0.05 percentage points more on an event that limited the digital euro, with a confidence interval that excludes effects one sixth the size of those documented for the ECB’s announcements of 2020 and 2021. The magnitude bound survives the pre-specified robustness checks, a wild cluster bootstrap, a conservative release-time alignment and a restriction to events that were contested and unscheduled, which argues against anticipation as the explanation. The release-time-aligned estimate of 0.15 to 0.17 is significant at the 10% level by the wild bootstrap under every benchmark and at the 5% level under a fully euro-denominated one, with permutation p -values between 0.05 and 0.12; we read it as some evidence of a small effect, not as a robust finding. On the events that set or signalled a design parameter, as opposed to events that only changed the probability of issuance, the estimate is

−0.004 percentage points with a confidence interval from −0.22 to 0.21: we detect no price effect of the design parameters. And the small positive slope that remains in the pooled estimate appears, if anything, larger for banks outside the euro area, which is consistent with a generic factor loading on deposit-funded banks rather than a digital-euro-specific response.

For financial stability the implication is direct. The concern that motivated the holding limit was deposit flight from banks. Markets did not price a large residual flight risk into euro area bank valuations during the legislative process; any effect on the most exposed banks appears modest relative to their response to the early announcements. The market did react to digital euro news: on the day the Eurosystem report was published in October 2020, deposit-funded banks underperformed by 0.85 percentage points per standard deviation of predetermined exposure. But that was news about whether the instrument would exist. News about how it would be designed, from the first cap signal in 2021 to the legislative texts of 2023 to 2026, did not generate a consistent or statistically detectable response of deposit-funded banks in the predicted direction. The negotiation record shows that the parameters were nonetheless intensely contested, with 309 documented meetings and the banking side accounting for the largest share. Whatever the banks were bargaining over did not generate a large detectable valuation response in equity markets.

Two questions remain open. First, whether the null extends to lending and deposit pricing, which equity prices do not observe; bank-level lending data around the design events would answer it. Second, whether the timing of releases whose publication time is not documented matters; aligning every event to its documented release circumstances raises the point estimate to 0.16 without changing the bound, and exact timestamps for the remaining thirteen releases would settle the question.

Data and code availability

All data used in this paper are public. Event documents come from EUR-Lex, the Legislative Observatory, the Council, the Eurogroup and the ECB (sources in Table 8); the negotiation record is transcribed from procedure file 2023/0212(COD). Bank exposure comes from the EBA EU-wide transparency exercises of 2020 and 2023, factor returns from the Kenneth French Data Library, and the deposit facility rate from the ECB Data Portal (series FM.D.U2.EUR.4F.KR.DFR.LEV). Share prices are from Yahoo Finance, whose terms govern redistribution; the package includes the scripts that retrieve them. The replication package contains the event coding, the negotiation record, the derived data, every estimation and figure script, a master script that reproduces all tables and figures, and the analysis plan with its dated SHA-256 freeze hashes: <https://github.com/Philip-Kroos/The-Price-of-a-Design-Parameter-Bank-Valuations-in-the-Digital-Euro-Legislative-Process>.

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A Codebook and full event list

A.1 Coding rules

Except for the two February 2024 events documented in Table 8, whose text was not available, each event is assigned from the contemporaneously available text of the act alone, before any price data is loaded:

- **Procedural**: one if the act contains no provision bearing on any design parameter.
- **Direction by group**, in $\{-1, 0, +1\}$, separately for banks, payment providers and device gatekeepers. Positive means the act raises the rent of that group relative to the status quo ante, by the logic of Propositions 3 to 5. A blank denotes an act with no provision bearing on that group.
- **Dose**: the quantitative parameter named in the act, where one is named.
- **Surprise**: none, low, medium or high, from the prior public record.
- **Justification**: the provision on which the coding rests, recorded in full.

The lobbying record is used only for the surprise dimension and never for direction, so that the treatment and the intensity measure remain separate. The coding file is hash-frozen; the hash is in the replication package.

A.2 Full event list

Table 8: Full event list, coding and estimation status

Date	ID	Banks	Prov.	Dev.	Treatment	Documented content
2023-06-28	E01	—	—		Coded	European Commission publishes the legislative proposal establishing the digital euro (COM(2023)0369): legal tender status, online and offline use, distribution by payment service providers, ECB may set holding limits (level not specified), no remuneration.
2023-10-19	E02	0	0		Coded	Announcement in Parliament that the proposal is referred to the ECON committee and associated committees.
2023-10-31	E03	—	0		Coded	ECB publishes its formal legal opinion on the proposal (CON/2023/0034), supporting the project and arguing for ECB discretion over holding limits.
2024-02-09	E04	0	0	0	Zero in pre-specified sample; excluded without look-ahead	ECON rapporteur (Berger) tables a first draft report on the proposal; the text could not be retrieved.
2024-02-20	E05	0	0		Coded	LIBE committee adopts its opinion, concerning privacy and data protection only.
2024-02-21	E06	0	0	0	Zero in pre-specified sample; excluded without look-ahead	Two sets of amendments to the ECON draft report are tabled. Their content could not be retrieved.
2024-11-13	E07	0	0		Coded	Formal resumption of the file after the European elections (procedural decision to continue business from the previous term).

Date	ID	Banks	Prov.	Dev.	Treatment	Documented content
2024-12-16	E08	0	0	0	Coded	Change of ECON rapporteur: Berger steps down, Navarrete Rojas (same political group) is appointed. The new rapporteur has no publicly documented position on the file at this date.
2025-07-03	E09	0	0		Coded	A new LIBE rapporteur for the opinion is appointed.
2025-07-08	E18				Excluded: not verified at a primary source	ECOFIN expresses backing for swift progress on the file.
2025-07-15	E10	0	0		Coded	LIBE committee adopts a new opinion, concerning privacy and data protection only.
2025-09-02	E26	+	+	0	Coded	The ECON rapporteur publishes a 27-page chapter in a yearbook, presented as a personal contribution, arguing that the digital euro is not needed and pointing to private European payment solutions (naming the European Payments Initiative) as the alternative.
2025-09-19	E19	0	0	0	Coded	Eurogroup ministers endorse a framework for how the ceiling on individual holding limits will be set and amended; no numeric limit is agreed.
2025-10-23	E23	—	—	0	Coded	European Council leaders call for accelerated progress on the digital euro; no design content.
2025-10-30	E11	—	—		Coded	ECB closes the preparation phase and announces the move to the next phase of the project, including preparation for possible issuance.
2025-11-03	E12	+	+	0	Coded	New ECON draft report: an offline digital euro is introduced immediately; an online digital euro is introduced only if the Commission, after an investigation, concludes that no suitable private pan-European payment solution exists; holding limits made more prescriptive, set by the Commission based on an ECB report.
2025-12-19	E22	0	—	—	Coded; merged with same-day event (E13)	Council agrees its negotiating position: ECB sets holding limits within an overall ceiling agreed by the Council, reviewed at least every two years; providers may not charge consumers for basic services; guaranteed access for providers to mobile device hardware and software; interchange and merchant fees capped for at least five years at levels of comparable payment means, then cost-based. On the same day four sets of amendments to the ECON draft report are tabled.
2025-12-19	E13	0	0	0	Coded; merged with same-day event (E22)	Four sets of amendments to the ECON draft report are tabled, after the deadline of 12 December 2025.
2026-03-05	E20	—	0	0	Coded	ECB publishes a call for payment service providers to express interest in a 12-month pilot planned for the second half of 2027.
2026-06-23	E14	+	—		Coded	ECON committee adopts its position: overall cap on holdings set by the Commission on the ECB's recommendation and reviewed every two years; legal persons may not hold digital euro except incoming payments for up to 24 hours; rollout phase of at least 24 months; merchant and inter-provider fees capped; offline payments free of charge.
2026-06-26	E15	0	0		Coded	The committee report is formally tabled for plenary; content identical to the committee position of 23 June.
2026-07-06	E16	0	0		Coded	Announcement in plenary of the committee's decision to open negotiations (procedural).

Date	ID	Banks	Prov.	Dev.	Treatment	Documented content
2026-07-09	E17	+	—		Coded	Plenary confirms the negotiating mandate on the basis of the committee text.
2026-07-13	E21				Excluded: not verified at a primary source	First trilogue between Parliament and Council.
2026-07-14	E24	—	0	0	Coded	ECB announces the selection of 36 payment service providers (banks and non-banks) from more than 50 applicants for the pilot.
2026-09-15	E25	—	0	0	Excluded: after end of factor sample	ECB calls on e-commerce and mobile-commerce merchants to join the pilot.

Notes: Direction is the coded design content for each group, assigned from the text of the act before any price data was loaded; blank cells denote acts with no provision affecting that group. The content column reproduces the documented content on which the coding rests.

B Proofs

Proof of Proposition 1. Write $S_i(L) = \sigma \int_0^{\bar{b}} \min(b, L) dF_i(b) = \sigma [\int_0^L b dF_i(b) + L(1 - F_i(L))]$. Differentiating under the integral,

$$S'_i(L) = \sigma [Lf_i(L) + (1 - F_i(L)) - Lf_i(L)] = \sigma [1 - F_i(L)] \geq 0,$$

and $S''_i(L) = -\sigma f_i(L) \leq 0$. Hence S_i is increasing and concave, and $S'_i(L) \rightarrow 0$ as $L \rightarrow \bar{b}$. $\square$

Proof of Proposition 2. The assumption is that F_i is first-order stochastically dominated by F_k , that is $F_i(x) \geq F_k(x)$ for all x . Then $1 - F_i(L) \leq 1 - F_k(L)$ pointwise, so $S'_i(L) \leq S'_k(L)$ for every L , and since $S_i(0) = S_k(0) = 0$, integration gives $S_i(L) \leq S_k(L)$. The level gap is

$$S_k(L) - S_i(L) = \sigma \int_0^L [F_i(x) - F_k(x)] dx,$$

which is weakly increasing in L , because the integrand is non-negative, and constant once L exceeds the upper end of both supports. The level gap is therefore not maximised at an interior value of L . What can peak in the interior is the difference in marginal effects, $S'_k(L) - S'_i(L) = \sigma [F_i(L) - F_k(L)]$, which is largest where the two distribution functions differ most. $\square$

Proof of Propositions 3 to 5. Bank rent is $\Pi_i^B = s[D_i - S_i(L) - \lambda C_i] + \phi_i$ with ϕ_i independent of L and λ by construction, since distribution income is set by the fee schedule rather than by the limit. Then $\partial \Pi_i^B / \partial L = -sS'_i(L) = -s\sigma[1 - F_i(L)] < 0$ whenever $F_i(L) < 1$. Differentiating with respect to λ gives $\partial \Pi_i^B / \partial \lambda = -sC_i \leq 0$, with strict inequality when corporate balances C_i are positive.

Provider rent is $\Pi_j^P = \min(m, \bar{f})V_j + a\rho_j rV^{\text{mobile}} - c_j$, which is weakly increasing in $\bar{f}$ and strictly increasing when the cap binds, with derivative V_j . It does not depend on L , so the fee channel and the quantity channel are orthogonal at first order.

Gatekeeper rent is $\Pi^G = (1 - a)rV^{\text{mobile}}$, so $\partial \Pi^G / \partial a = -rV^{\text{mobile}} < 0$. Provider rent contains $a\rho_j rV^{\text{mobile}}$, so $\partial \Pi_j^P / \partial a = \rho_j rV^{\text{mobile}} \geq 0$, strictly positive when $\rho_j > 0$. The gatekeeper loss follows from the model; the provider gain rests on the assumption that providers capture part of the dissipated

rent. □

Remark on the sign pattern. Because L , $\bar{f}$ and a enter different rent functions, an act that lowers L and lowers $\bar{f}$ raises Π^B and lowers Π^P at the same time. A confounding factor that generates the same pattern would have to be correlated with bank exposure and provider exposure with opposite signs on the same trading day, which is the sense in which the sign pattern is identifying.

C The negotiation record: construction

Source. The transparency section of procedure file 2023/0212(COD) on the European Parliament’s Legislative Observatory. Publication is mandatory for members acting as rapporteur, shadow rapporteur or committee chair, and voluntary for other members. We transcribe the mandatory entries only, so every count in the paper refers to members acting in a reporting capacity.

Transcription. Each record yields four fields: date, member, role and organisation. Where a single record lists several organisations, we keep the record as one meeting and assign it to the first named organisation, which understates the presence of the others. Where the same organisation appears twice on one date with different members, both are counted, because the object of interest is contact intensity rather than the number of distinct organisations.

Categories. Organisations are assigned to seven categories: banks, banking associations, payment providers and schemes, large technology firms, retailers, civil society and consumer organisations, and other. The residual category is dominated by central banks, finance ministries and permanent representations, which are public bodies and are excluded from every measure of private interest. The assignment is deterministic given the name and is recorded in the replication package, so it can be re-run under a different classification.

Intensity. For each event we count meetings in the thirty calendar days before the event date, in total and by category. Thirty days is chosen because it spans the interval between committee meetings; counts over seven and sixty days are in the replication package, and across the 22 estimation dates their rank correlations with the thirty-day count are 0.63 and 0.91.

Known limitations. The record captures meetings, not their content, duration or outcome. It captures contact with legislators, not with the Commission or the ECB, each of which maintains a separate register we do not use. And it is a lower bound on contact, since informal exchange is not recorded. None of this affects the use we make of it, which is descriptive and comparative across events.

D Data construction

D.1 Firm universe

Table 9: Firm universe

Group	Country	Firm and reason for inclusion
bank	DE	Deutsche Bank: large deposit base; 12 documented meetings on the file
bank	DE	Commerzbank: retail deposit funded
bank	FR	BNP Paribas: 5 documented meetings
bank	FR	Société Générale: documented meetings
bank	FR	Crédit Agricole: documented meetings
bank	ES	Banco Santander: 7 documented meetings
bank	ES	BBVA: documented meetings
bank	ES	CaixaBank: documented meetings; high retail deposit share
bank	ES	Banco Sabadell: retail deposit funded
bank	ES	Bankinter: retail deposit funded
bank	ES	Unicaja: retail deposit funded
bank	IT	Intesa Sanpaolo: documented meetings
bank	IT	UniCredit: documented meetings
bank	IT	Banco BPM: retail deposit funded
bank	IT	BPER Banca: retail deposit funded
bank	IT	Mediobanca: lower retail deposit share; useful low-exposure bank
bank	NL	ING Groep: documented meetings; very high deposit share
bank	NL	ABN AMRO: high retail deposit share
bank	BE	KBC Group: retail deposit funded
bank	AT	Erste Group: documented meeting
bank	AT	Raiffeisen Bank International: retail deposit funded
bank	AT	BAWAG Group: retail deposit funded
bank	IE	Bank of Ireland: retail deposit funded
bank	IE	AIB Group: retail deposit funded
bank	GR	National Bank of Greece: retail deposit funded
bank	GR	Eurobank: retail deposit funded
bank	GR	Alpha Services: retail deposit funded
bank	GR	Piraeus Financial: retail deposit funded
bank	PT	Banco Comercial Português: retail deposit funded
bank	CY	Bank of Cyprus: retail deposit funded
payments	US	Mastercard: card network; documented meetings
payments	US	Visa: card network; documented meeting
payments	US	American Express: card network; documented meeting
payments	US	PayPal: digital wallet incumbent
payments	IT	Nexi: euro area acquirer and processor; documented meeting
payments	FR	Worldline: euro area acquirer and processor
payments	NL	Adyen: euro area acquirer
payments	UK	Wise: cross-border payments; documented meeting
payments	US	Fiserv: merchant acquiring
payments	US	Global Payments: merchant acquiring
device	US	Apple: device gatekeeper directly targeted by the NFC access provisions; documented meetings
device	US	Alphabet: device and wallet operator
bigtech	US	Amazon: 8 documented meetings; e-commerce acceptance side
bigtech	US	Meta: documented meeting
control bank	SE	Swedbank: non-euro bank with similar deposit structure
control bank	SE	Svenska Handelsbanken: non-euro control
control bank	DK	Danske Bank: non-euro control
control bank	NO	DNB Bank: non-euro control
control bank	UK	Lloyds Banking Group: non-euro control with high retail deposit share

Group	Country	Firm and reason for inclusion
control bank	UK	NatWest Group: non-euro control
control bank	CH	UBS Group: non-euro control; documented meeting

D.2 Robustness of the main specification

Table 10: Robustness of the main bank-exposure estimate

Variation	Estimate	<i>t</i>
Baseline: one day, three factors, common 250-day estimation window	0.051	0.92
Three-day window	−0.006	−0.08
Common estimation window of 120 days	0.061	1.11
Common estimation window of 500 days	0.050	0.92
Excluding the March 2023 banking turmoil from estimation	0.051	0.92
Market model on the STOXX Europe 600	0.055	0.97
EUR market factor with FF size and value, one day	0.042	0.75
EUR market factor with FF size and value, two days	0.115	1.21
EUR market factor with FF size and value, release-time aligned	0.147	1.61
EUR market factor with FF size and value, design events	−0.003	−0.02
Euro market model in excess returns, one day	0.054	0.95
Euro market model in excess returns, release-time aligned	0.167	2.04
Euro market model in excess returns, design events	−0.001	−0.01
Pre-event estimation window per event, days −260 to −11	0.099	1.83
F3: limiting events only (direction +1)	0.005	0.05
F3: expanding events only (direction −1)	0.096	0.92
F1: exposure slope on the four procedural dates	0.040	0.55
Council position dated 17 December 2025 (working document)	0.049	0.87
Draft report dated 5 November 2025 (committee presentation)	−0.014	−0.20

Notes: Percentage points of abnormal return per standard deviation of the deposit share per unit of coded direction, with the group-level terms and firm and event fixed effects of the main specification; *t* from leave-one-firm-out jackknife standard errors. In the baseline, factor loadings are estimated once per firm on the most recent 250 trading days that lie at least ten days from any event, which in this sample falls after March 2023; excluding the banking turmoil therefore leaves the estimate unchanged. The conventional alternative estimates loadings separately for each event on days −260 to −11 before it. The four-factor model with a European banks index was planned but not estimated because the index series could not be obtained. Under F3 both split coefficients are predicted to be positive; they are 0.005 and 0.096 and neither is distinguishable from zero. The two re-dated rows move one event to its alternative date and re-estimate both stages. Under F1 the slope on procedural dates is predicted to be zero; it is estimated jointly with the main term, which stays at 0.050, and its permutation *p*-value is 0.73.

D.3 Returns and factors

Daily returns are log changes of dividend- and split-adjusted closing prices in the currency of listing. Factor returns are the European daily three-factor series from the Fama and French data library, which covers the sample from June 2019 to 31 July 2026. These factors are computed in US dollars, so the pre-specified first stage combines local-currency returns with a dollar-denominated market factor. We therefore also report a first stage in which the market factor is replaced by the euro-denominated return on the STOXX Europe 600 while the dollar-based size and value factors are kept, which reduces but

does not eliminate the currency mismatch; and a fully euro-denominated market model in excess returns, with the ECB deposit facility rate as the risk-free rate. Because every regression includes a firm-specific intercept, using raw rather than excess returns affects only the intercept to first order; the daily risk-free rate varies by less than 0.02 percentage points over the sample. Prices are daily closes adjusted for dividends and splits, for June 2019 to September 2026. A firm-event observation is retained when at least 120 clean trading days are available before that event; firms that enter the listed sample later contribute only to subsequent events.

D.4 Exposure

Bank exposure uses the EBA transparency exercise reference date closest to and before June 2023. The planned provider exposure measure, the euro area revenue share from segment reporting, and the planned concentration measure from ECB structural financial indicators were not assembled; providers enter with a group indicator. All implemented exposure measures are fixed at their pre-proposal values and never updated, so that exposure cannot respond to the events.

E Pre-specification and deviations

What was fixed, and how. The analysis plan is not held in an external registry. It is a written plan kept in the replication archive and frozen with SHA-256 hashes at dated points, each before the data it governs were used: the event coding and the firm universe before any return was computed; the definition of the bank exposure measure before any estimate with it; the coding of the 2020 and 2021 events before any return from the earlier price file was computed; and the historical exposure measure before any estimate with it. The hashes and timestamps are in the archive. We therefore describe the analysis as pre-specified rather than pre-registered.

Deviations. Table 11 lists every departure from the plan, with its reason and the point at which it was decided.

Every analysis added after the first estimates is labelled as such in the text and reported alongside, never in place of, the pre-specified estimate.

Table 11: Departures from the pre-specified plan

Planned	Implemented	Reason	Decided
Three-factor model for all firms	Market model on the S&P 500 for US-listed firms	European factors are not a benchmark for US equities	Before any abnormal return
Falsification test assigning the bank direction to non-euro banks (binary design)	Replaced by placebo dates	Collinear with event fixed effects in the binary design	After first estimation
Continuous provider exposure (euro area revenue share)	Group indicator	Segment revenue data not assembled	Before estimation
Market-power split (deposit concentration)	Not implemented	Concentration data not assembled	Before estimation
Provider-role split (distributing vs acquiring)	Not implemented	Provider-level role and revenue data not assembled	Before estimation
F4: placebo dates from the same weeks as the events	Placebo dates from non-event trading days since January 2023, at least ten trading days from any event	Same-week days fall inside the exclusion band of the first stage and overlap the event windows	After first estimation
Within-group test: selected vs non-selected pilot providers	Not implemented	Usable firm-level list of selected providers not available	Before estimation
Coding of the February 2024 draft and amendments as zero	Additionally excluded in a no-look-ahead specification	Original coding relied partly on the later fate of the draft	After review
Pooled design and implementation events	Additionally separated	Title question concerns design events	After review
One-day window with documented event days	Additionally release-time aligned	Publication time undocumented for 13 events	After review
Exposure measured in 2023 for all events	December 2019 exposure for the 2020–21 events	2023 measure not predetermined for early events	After review, before estimation
Pre-specified first stage mixes local-currency returns with dollar factors	Additionally, a euro market factor with FF size and value, and a euro market model in excess returns	Fama–French European factors are computed in US dollars	After review
Two-day windows summed over available days	Windows require a return on every day	Coding error; two-day results recomputed	After review
Four-factor model with a European banks index	Not estimated	Index series could not be obtained	Before estimation
Romano–Wolf with permutation draws	Romano–Wolf with placebo-date draws	Permutation within a binary group is degenerate	After first estimation